

ISC Class XII Mathematics

Chapter 13 Probability

69 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Topics Covered

- I. Addition Theorem of Probability (Types 1–11)
- II. Conditional Probability (Types 12–20)
- III. Multiplication Theorem & Independent Events (Types 21–43)
- IV. Total Probability (Types 44–51)
- V. Bayes' Theorem (Types 52–57)
- VI. Probability Distributions (Types 58–63)
- VII. Mean (Expectation) of a Probability Distribution (Types 64–69)

Contents

1	I. Addition Theorem of Probability	7
2	II. Conditional Probability	17
3	III. Multiplication Theorem & Independent Events	25
4	IV. Total Probability	47
5	V. Bayes' Theorem	54
6	VI. Probability Distributions	61
7	VII. Mean (Expectation) of a Probability Distribution	67

Index of Types

Type	Method / Pattern	★	Source(s)
1	Addition theorem – mutually exclusive events	–	Ex 13.1 Q1,2
2	Multi-part complement & union computations	Yes	Ex 13.1 Q3,4,5
3	Algebraic use of addition formula for unknown	–	Ex 13.1 Q6,7,8,14
4	Exhaustive events – ratio substitution technique	–	Ex 13.1 Q9
5	Classical probability – dice “or” events	–	Ex 13.1 Q10,20,21
6	Feasibility check via $P(A \cup B) \leq 1$	–	Ex 13.1 Q11
7	Algebra: atleast-one $\rightarrow$ both, from addition thm	–	Ex 13.1 Q12
8	Classical probability – cards, union of events	Yes	Ex 13.1 Q17,18,19
9	Set-overlap word problem (subjects opted)	–	Ex 13.1 Q15
10	Union probability via direct counting	–	Ex 13.1 Q13,16
11	Combinatorial union – “both X or both Y”	–	Ex 13.1 Q22,23,24

Type	Method / Pattern	★	Source(s)
12	Conditional probability – direct formula	–	Ex 13.2 Q2,3,4,5,6
13	Conditional probability – $P(B) = 0$ & symmetry	–	Ex 13.2 Q1
14	Conditional probability – reduced sample space	–	Ex 13.2 Q7,8,9,12
15	Conditional probability – family/reduced-space classic	– Yes	Ex 13.2 Q10,11
16	Conditional probability – overlapping card events	–	Ex 13.2 Q13
17	Conditional probability – algebraic dice condition	–	Ex 13.2 Q14
18	Conditional probability – percentage word problems	–	Ex 13.2 Q15,16
19	Conditional probability with combinatorics	–	Ex 13.2 Q17
20	Conditional probability – arrangement logic puzzle	–	Ex 13.2 Q18
21	Foundational facts – exclusive vs independent	–	Ex 13.3 Q1
22	Classical probability by direct counting (dice)	–	Ex 13.3 Q2,27,28
23	Testing independence from given probabilities	–	Ex 13.3 Q3,4(i,ii),5,6(i)
24	Independent events – union/intersection/complement	– –	Ex 13.3 Q4(iii,v,vi)
25	Independent events – reverse-solve unknown p	–	Ex 13.3 Q4(iv),6(ii)
26	“At least one” success – independent multi-trial	Yes	Ex 13.3 Q4(vii),42
27	Classify: independent / exclusive / neither	Yes	Ex 13.3 Q7,8
28	Combinatorics – hypergeometric team selection	–	Ex 13.3 Q9(i),17(i)
29	Sequential draw w/o replacement – mixed outcome	–	Ex 13.3 Q9(ii),17(ii)
30	With-replacement repeated draw probability	–	Ex 13.3 Q10,18
31	“At least one of several” independent successes	–	Ex 13.3 Q11,32,34,44
32	Sequential w/o replacement – “all same” / “none of a kind”	Yes	Ex 13.3 Q12,13,15,19,37,38,39,41
33	With-replacement – repeated identical draw	–	Ex 13.3 Q12(ii),13(ii)
34	Combinatorics – “both A or both B” union	–	Ex 13.3 Q14,23,25

Type	Method / Pattern	★	Source(s)
35	Sequential fixed-order specific sequence	–	Ex 13.3 Q16,39
36	Combinatorics – number-property pairs (prime/multiple)	–	Ex 13.3 Q17,20,36(i)
37	Independent draws from two sources – same colour	–	Ex 13.3 Q22,26,40
38	Combinatorics – composite category selection	–	Ex 13.3 Q24
39	Multi-agent independent success exactly- k /atleast- k	–	Ex 13.3 Q32,33
40	Games played alternately – geometric series	Yes	Ex 13.3 Q43,45–52
41	Algebra – recover $P(A)$, $P(B)$ from independence eqns	–	Ex 13.3 Q53,54
42	Venn diagram – direct-reading probability	–	Ex 13.3 Q55
43	Reverse-conditional via multiplication theorem	–	Ex 13.3 Q29,30
44	Total probability – two-source selection	–	Ex 13.4 Q1,2,7
45	Total probability – demographic word problem	–	Ex 13.4 Q3,9
46	Total probability – conditional business scenario	–	Ex 13.4 Q4
47	Total probability – sequential transfer between bags	–	Ex 13.4 Q5
48	Total probability – “return + add same colour” scheme	–	Ex 13.4 Q6
49	Total probability – three or more sources	–	Ex 13.4 Q8,9
50	Total probability with combinatorics – multi-ball draw	–	Ex 13.4 Q10
51	Total probability – transfer then combinatorial draw	–	Ex 13.4 Q11
52	Bayes’ theorem – standard two-source reverse probability	Yes	Ex 13.5 Q1,2,6,7,9,10,19
53	Bayes’ theorem – weights from population counts	–	Ex 13.5 Q3,4,5,8,20–23
54	Bayes’ theorem with combinatorics (both balls same)	–	Ex 13.5 Q11–14,17,28–31,33,34
55	Bayes’ theorem – sequential transfer then Bayes	–	Ex 13.5 Q15,16
56	Bayes’ theorem – special coin/box puzzles	Yes	Ex 13.5 Q18,25,26
57	Bayes’ theorem – comprehensive multi-agent	–	Ex 13.5 Q24,27,32,35,36
58	Verifying a distribution & finding unknown constant	–	Ex 13.6 Q1,2,3

Type	Method / Pattern	★	Source(s)
59	Piecewise-defined distribution, find k	–	Ex 13.6 Q3,4
60	Distribution from a simple experiment (coin/die)	–	Ex 13.6 Q5,6
61	Distribution via combinatorics (balls/cards w/o replacement)	–	Ex 13.6 Q7,8,9,10–14
62	Distribution – table-based custom experiment	Yes	Ex 13.6 Q16
63	Distribution – dice-difference	–	Ex 13.6 Q15
64	Mean – simple discrete distribution, direct table	–	Ex 13.7 Q3,4,5
65	Fair-game / expectation-of-winnings analysis	–	Ex 13.7 Q1,2
66	Mean – unknown constant k found first	–	Ex 13.7 Q6,7,8,9,10
67	Mean – combinatorics-based distribution	–	Ex 13.7 Q11–18,22,23,24
68	Mean – order statistic (larger/largest of several)	–	Ex 13.7 Q19,20,21
69	Mean – repeated-trial scoring game	Yes	Ex 13.7 Q25

Formula Sheet**Addition Theorem**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

General union formula for any two events

Mutually Exclusive Events

$$P(A \cup B) = P(A) + P(B)$$

Valid only when $A \cap B = \phi$

Complement Rule

$$P(A') = 1 - P(A)$$

A' = event A does not occur

Difference of Events

$$P(A \cap B') = P(A) - P(A \cap B)$$

Probability A occurs but B does not

De Morgan (Neither)

$$P(A' \cap B') = 1 - P(A \cup B)$$

Probability neither A nor B occurs

De Morgan (Not Both)

$$P(A' \cup B') = 1 - P(A \cap B)$$

Probability at least one of A, B fails

Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Probability of A given B has occurred

Multiplication Theorem

$$P(A \cap B) = P(A) \cdot P(B | A) = P(B) \cdot P(A | B)$$

Compound probability of both A and B

Independent Events

$$P(A \cap B) = P(A) \cdot P(B)$$

Occurrence of one does not affect the other; equivalently $P(A|B) = P(A)$

At Least One Success

$$P(\text{atleast one}) = 1 - P(\text{none})$$

Complement is easier to compute directly

Combinations

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Number of ways to choose r objects from n , order irrelevant

Law of Total Probability

$$P(A) = \sum_{i=1}^n P(E_i) \cdot P(A | E_i)$$

$E_1, \dots, E_n$ mutually exclusive and exhaustive “sources”

Bayes’ Theorem

$$P(E_i | A) = \frac{P(E_i) \cdot P(A | E_i)}{\sum_j P(E_j) \cdot P(A | E_j)}$$

Reverses a conditional – probability a specific source caused the observed outcome

Probability Distribution

$$\sum_i P(X = x_i) = 1$$

Every valid probability distribution’s probabilities must sum to exactly 1

Mean (Expectation)

$$\mu = E(X) = \sum_i p_i x_i$$

Long-run average value of the random variable X

I. Addition Theorem of Probability

Type 1

Two events are given as *mutually exclusive* (they cannot both happen at once, so $A \cap B = \phi$), and their individual probabilities are given. Recognise the type from the phrase “mutually exclusive events” combined with a request for $P(A \text{ or } B)$.

Formula Used: $P(A \cup B) = P(A) + P(B)$ (when $A \cap B = \phi$)

Source: Exercise 13.1, Q1

Given: $P(A) = \frac{3}{5}$, $P(B) = \frac{1}{5}$; A and B are mutually exclusive events.

Solution:

Since A and B are mutually exclusive, they share no common outcomes, so $P(A \cap B) = 0$ – there is nothing to subtract in the addition formula.

$$P(A \cup B) = P(A) + P(B) = \frac{3}{5} + \frac{1}{5} = \frac{4}{5}$$

We simply add the two probabilities because no outcome is being double-counted.

Final Answer

$$P(A \text{ or } B) = \frac{4}{5}$$

Common Mistake: Students often apply the full formula $P(A) + P(B) - P(A \cap B)$ and then guess a nonzero value for $P(A \cap B)$ instead of recognising that “mutually exclusive” forces it to be exactly 0.

Quick Recall: Mutually exclusive means no overlap – just add the two probabilities directly.

Type 2

A pair of events is given (sometimes explicitly mutually exclusive, sometimes a general pair with an intersection value given), and several related probabilities are asked together: complements, union, intersection, and combinations like $P(A \cap B')$ or $P(A' \cap B')$. Recognise this from the “find (i)...(vi)” multi-part format.

★ **Board Favorite** – Multi-part complement/union questions like this test several formulas at once and appear frequently as 4–6 mark ISC questions.

Formula Used: $P(A') = 1 - P(A)$

Formula Used: $P(A \cap B') = P(A) - P(A \cap B)$

Formula Used: $P(A' \cap B') = 1 - P(A \cup B)$

Source: Exercise 13.1, Q3

Given: A and B are mutually exclusive events, $P(A) = 0.35$, $P(B) = 0.45$. Find (i) $P(A')$ (ii) $P(B')$ (iii) $P(A \cup B)$ (iv) $P(A \cap B)$ (v) $P(A \cap B')$ (vi) $P(A' \cap B')$.

Solution:

Note: This is the **mutually exclusive** sub-case (2a): since $A \cap B = \phi$, several answers collapse to simple forms. Compare with the general sub-case (2b) below, worked from Q5, where $A \cap B \neq \phi$.

- (i) Complement rule: $P(A') = 1 - P(A) = 1 - 0.35 = 0.65$.
- (ii) Similarly, $P(B') = 1 - 0.45 = 0.55$.
- (iii) Mutually exclusive events simply add: $P(A \cup B) = 0.35 + 0.45 = 0.8$.
- (iv) Mutually exclusive means no overlap, so $P(A \cap B) = 0$.
- (v) Since $A \cap B = \phi$, every outcome of A already avoids B , so $A \cap B' = A$: $P(A \cap B') = P(A) - P(A \cap B) = 0.35 - 0 = 0.35$.
- (vi) By De Morgan, neither A nor B occurs is the complement of their union: $P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.8 = 0.2$.

Final Answer

(i) 0.65 (ii) 0.55 (iii) 0.8 (iv) 0 (v) 0.35 (vi) 0.2

Common Mistake: Confusing $P(A \cap B')$ with $P(A') \cap P(B)$, or forgetting that for mutually exclusive events $A \cap B'$ is the same set as A itself.

Alternate Board-Style Example

Source: Exercise 13.1, Q5

Given:

Note: Sub-case (2b): general (non-exclusive) pair – here $P(A \cap B)$ is given directly and is nonzero, so the shortcuts from (2a) do not apply; the general formulas must be used instead.

A and B are events such that $P(A) = 0.42$, $P(B) = 0.48$, $P(A \text{ and } B) = 0.16$. Determine (i) $P(\text{not } A)$ (ii) $P(\text{not } B)$ (iii) $P(A \text{ or } B)$.

Solution:

(i) $P(\text{not } A) = 1 - 0.42 = 0.58$.

(ii) $P(\text{not } B) = 1 - 0.48 = 0.52$.

(iii) Here $A \cap B \neq \phi$, so the intersection genuinely must be subtracted: $P(A \text{ or } B) = P(A) + P(B) - P(A \cap B) = 0.42 + 0.48 - 0.16 = 0.74$.

Final Answer

(i) 0.58 (ii) 0.52 (iii) 0.74

Quick Recall: Complement always flips with $1 -$; for union, only subtract the intersection if it is genuinely nonzero – mutually exclusive events never need that subtraction.

Type 3

Three related probabilities are given (two individual, one combined – union or intersection) and one probability is missing. Recognise this from “find $P(F)$ ” style questions where you must isolate the unknown algebraically inside the addition formula.

Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Source: Exercise 13.1, Q7

Given: E and F are events with $P(E) = 0.60$, $P(E \text{ or } F) = 0.85$, $P(E \text{ and } F) = 0.42$. Find $P(F)$.

Solution:

Substitute the known values into the addition formula, leaving $P(F)$ as the only unknown.

$$0.85 = 0.60 + P(F) - 0.42$$

Combine the known terms on the right first, so the equation becomes a one-step solve.

$$0.85 = 0.18 + P(F) \implies P(F) = 0.85 - 0.18 = 0.67$$

Final Answer

$P(F) = 0.67$

Common Mistake: Sign errors when rearranging – students forget the $-P(E \cap F)$ term is being added back (i.e. subtracting a negative) when isolating $P(F)$.

Quick Recall: Plug everything you know into $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and solve for the one missing piece like ordinary algebra.

Type 4

Three (or more) events are mutually exclusive *and* exhaustive (together they cover the whole sample space, so their probabilities sum to 1), and their probabilities are related to each other by ratios (e.g. $P(B) = \frac{3}{2}P(A)$). Recognise this from “mutually exclusive and exhaustive” combined with probabilities expressed as multiples of one another.

Formula Used: $P(A) + P(B) + P(C) = 1$ (mutually exclusive and exhaustive)

Source: Exercise 13.1, Q9

Given: A, B, C are mutually exclusive and exhaustive events with $P(B) = \frac{3}{2}P(A)$ and $P(C) = \frac{1}{2}P(B)$. Find $P(A)$.

Solution:

Let $P(A) = x$, so every other probability can be written in terms of the single unknown x .

$$P(B) = \frac{3}{2}x, \quad P(C) = \frac{1}{2} \cdot \frac{3}{2}x = \frac{3}{4}x$$

Because the three events are exhaustive, their probabilities must add to exactly 1 (someone one of them must happen).

$$x + \frac{3}{2}x + \frac{3}{4}x = 1$$

Combine over a common denominator of 4 to solve for x cleanly.

$$\frac{4x + 6x + 3x}{4} = 1 \implies \frac{13x}{4} = 1 \implies x = \frac{4}{13}$$

Final Answer

$$P(A) = \frac{4}{13}$$

Common Mistake: Forgetting to convert $P(C)$ in terms of x correctly – since $P(C)$ is defined relative to $P(B)$, not $P(A)$, it must be substituted in two steps.

Quick Recall: When probabilities are given as ratios of each other, let the base one be x , rewrite everything else in terms of x , and use “exhaustive $\Rightarrow$ sum = 1”.

Type 5

A single random experiment (usually dice) has two events connected by “or”, and the two events overlap (some outcomes satisfy both). Recognise this from a compound condition like “an odd number on the first die *or* a total of 8”.

Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Source: Exercise 13.1, Q20

Given: Two dice are thrown once. Find the probability of getting an odd number on the first die *or* a total of 8.

Solution:

List the sample space size first: two dice give $6 \times 6 = 36$ equally likely outcomes. Let A = odd number on first die. Half of the first-die values are odd (1,3,5), so A has $3 \times 6 = 18$ outcomes, giving $P(A) = \frac{18}{36}$. Let B = total is 8: the pairs are (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), giving $P(B) = \frac{5}{36}$. Now find the overlap $A \cap B$ – outcomes with an odd first die *and* total 8: from the list above, (3, 5) and (5, 3) qualify, so $P(A \cap B) = \frac{2}{36}$.

$$P(A \cup B) = \frac{18}{36} + \frac{5}{36} - \frac{2}{36} = \frac{21}{36} = \frac{7}{12}$$

Final Answer

$$P(A \text{ or } B) = \frac{7}{12}$$

Common Mistake: Forgetting to subtract the overlap – treating the two events as if they were mutually exclusive when some outcomes (like (3, 5)) actually satisfy both conditions.

Quick Recall: Whenever “or” connects two dice conditions, check carefully whether any outcome satisfies both before deciding if you need to subtract an overlap.

Type 6

Two probabilities are given along with a bound on their intersection (e.g. “atmost 0.3”), and the question asks whether a certain value for the other event’s probability is *possible*. Recognise this from the phrase “is it possible that...”.

Formula Used: $P(A \cup B) \leq 1 \implies P(A) + P(B) - P(A \cap B) \leq 1$

Source: Exercise 13.1, Q11

Given: A and B are two candidates seeking admission. $P(A \text{ selected}) = 0.5$ and $P(\text{both selected})$ is atmost 0.3. Is it possible that $P(B \text{ selected}) = 0.7$?

Solution:

Since probability of a union can never exceed 1, use that inequality as the test.

$$P(A) + P(B) - P(A \cap B) \leq 1 \implies 0.5 + P(B) \leq 1 + P(A \cap B)$$

Rearrange to isolate $P(B)$, using the fact that $P(A \cap B) \leq 0.3$ is the loosest (most generous) bound allowed.

$$P(B) \leq 1 - 0.5 + 0.3 = 0.8$$

Since the proposed value 0.7 satisfies $0.7 \leq 0.8$, it does not violate the constraint, so it is logically valid.

Final Answer

Yes, $P(B) = 0.7$ is possible.

Common Mistake: Concluding “impossible” just because the numbers look tight, without actually deriving the correct upper bound from $P(A \cup B) \leq 1$.

Quick Recall: To check if a probability value is “possible”, derive the maximum allowed value from $P(A \cup B) \leq 1$ and compare.

Type 7

The probability that *at least one* of two events occurs is given (often as a fraction), along with individual or complement probabilities, and the intersection (“both”) is the unknown. Recognise this from “atleast one of the two events” combined with a request for “both events”.

Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Source: Exercise 13.1, Q12

Given: $P(A) = \frac{2}{3}$, $P(\text{not } B) = \frac{5}{9}$, and $P(\text{atleast one of the two events}) = \frac{4}{5}$. Find $P(\text{success in both}) = P(A \cap B)$.

Solution:

First convert the complement to $P(B)$, since the addition formula needs $P(B)$ directly.

$$P(B) = 1 - \frac{5}{9} = \frac{4}{9}$$

“Atleast one” is exactly $P(A \cup B)$, so substitute everything into the addition formula and solve for the only unknown, $P(A \cap B)$.

$$\frac{4}{5} = \frac{2}{3} + \frac{4}{9} - P(A \cap B)$$

Find a common denominator (45) to combine the right-hand terms before isolating $P(A \cap B)$.

$$\frac{4}{5} = \frac{30}{45} + \frac{20}{45} - P(A \cap B) = \frac{50}{45} - P(A \cap B)$$

$$P(A \cap B) = \frac{50}{45} - \frac{36}{45} = \frac{14}{45}$$

Final Answer

$$P(\text{both events}) = \frac{14}{45}$$

Common Mistake: Forgetting to convert $P(\text{not } B)$ to $P(B)$ before substituting – plugging the complement in directly gives a completely wrong equation.

Quick Recall: “Atleast one” always means $P(A \cup B)$ – convert any complements first, then solve the addition formula for whichever piece is missing.

Type 8

A single card is drawn from a standard 52-card deck, and the event is a union of two overlapping card categories (e.g. “diamond or king”, “neither king nor red”). Recognise this from classic card-probability phrasing with “or”, “either...or...both”, or “neither...nor”.

★ **Board Favorite** – Card-probability union/complement questions are one of the most consistently tested patterns in ISC Probability across years.

Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Source: Exercise 13.1, Q17

Given: A card is drawn from a well-shuffled pack of 52 playing cards. Find the probability that it is either a diamond or a king or both.

Solution:

There are 13 diamonds, so $P(\text{diamond}) = \frac{13}{52}$. There are 4 kings, so $P(\text{king}) = \frac{4}{52}$. The overlap is the king of diamonds – exactly 1 card belongs to both categories, so $P(\text{diamond} \cap \text{king}) = \frac{1}{52}$.

$$P(\text{diamond or king}) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

Final Answer

$$P = \frac{4}{13}$$

Common Mistake: Forgetting the king of diamonds is counted in both the 13 diamonds and the 4 kings, and so double-counting it by not subtracting the overlap.

Alternate Board-Style Example

Source: Exercise 13.1, Q19

Given: A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is neither a king nor a red card.

Solution:

It is easier to find the complement first: $P(\text{king or red})$, then subtract from 1. There are 4 kings and 26 red cards; among the 4 kings, 2 (the king of hearts and king of diamonds) are also red, so the overlap is 2.

$$P(\text{king or red}) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{28}{52}$$

Neither king nor red is the complement of this union.

$$P(\text{neither}) = 1 - \frac{28}{52} = \frac{24}{52} = \frac{6}{13}$$

Final Answer

$$P = \frac{6}{13}$$

Quick Recall: For card unions, count each category, find the genuine overlap carefully (colour/suit/rank clashes), then apply the addition formula – or use the complement when “neither” is asked.

Type 9

A word problem describes two overlapping categories within a group of people (e.g. subjects opted for), giving totals for each category and the overlap; the question asks for various combinations. Recognise this from “opted for X, opted for Y, opted for both”.

Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Formula Used: $P(A' \cap B') = 1 - P(A \cup B)$

Source: Exercise 13.1, Q15

Given: In a class of 60 students, 30 opted for Mathematics, 32 opted for Biology, and 24 opted for both. One student is selected at random. Find (i) $P(\text{Maths or Biology})$ (ii) $P(\text{neither})$ (iii) $P(\text{Maths but not Biology})$.

Solution:

Convert the raw counts to probabilities over the total of 60 students first: $P(M) = \frac{30}{60}$, $P(B) = \frac{32}{60}$, $P(M \cap B) = \frac{24}{60}$. (i)

$$P(M \cup B) = \frac{30}{60} + \frac{32}{60} - \frac{24}{60} = \frac{38}{60} = \frac{19}{30}$$

(ii) Neither subject is the complement of the union.

$$P(\text{neither}) = 1 - \frac{19}{30} = \frac{11}{30}$$

(iii) “Maths but not Biology” means in M but excluded from the overlap with B .

$$P(M \cap B') = P(M) - P(M \cap B) = \frac{30}{60} - \frac{24}{60} = \frac{6}{60} = \frac{1}{10}$$

Final Answer

(i) $\frac{19}{30}$ (ii) $\frac{11}{30}$ (iii) $\frac{1}{10}$

Common Mistake: Adding 30+32 and forgetting to subtract the 24 who are counted in both groups, which inflates the “Maths or Biology” count beyond the class size.

Quick Recall: Turn word-problem counts into probabilities by dividing by the total, then treat it exactly like a standard addition-theorem question.

Type 10

A collection of items (tickets, or items in a bag) is described with overlapping properties (e.g. numbers divisible by 3 or 7; rusted or a bolt), and the probability of the union is found using direct item counts rather than card/dice knowledge. Recognise this from “how many items have property X, how many have property Y”.

Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Source: Exercise 13.1, Q16

Given: A bag contains 150 nuts and 50 bolts. Half the bolts and half the nuts are rusted. One item is drawn at random. Find the probability that it is either rusted or a bolt.

Solution:

Total items = 150 + 50 = 200. Rusted items = half of nuts + half of bolts = 75 + 25 = 100, so $P(\text{rusted}) = \frac{100}{200}$. Bolts total 50, so $P(\text{bolt}) = \frac{50}{200}$. The overlap “rusted and bolt” is exactly the rusted bolts, which is half of the 50 bolts, i.e. 25: $P(\text{rusted} \cap \text{bolt}) = \frac{25}{200}$.

$$P(\text{rusted or bolt}) = \frac{100}{200} + \frac{50}{200} - \frac{25}{200} = \frac{125}{200} = \frac{5}{8}$$

Final Answer

$$P = \frac{5}{8}$$

Common Mistake: Treating “rusted” and “bolt” as mutually exclusive and simply adding $100 + 50$, ignoring that some bolts are themselves rusted.

Quick Recall: Break composite items into their overlapping sub-groups first (rusted nuts, rusted bolts) before applying the addition formula.

Type 11

Two cards (or two fruits, two items) are drawn together, and the question asks for the probability of a compound “either both-X or both-Y” event, requiring combinations (nC_r) rather than simple counts. Recognise this from “both are aces or both are black”, “both apples or both good”.

Formula Used: $P(\text{both A or both B}) = \frac{{}^kC_2}{{}^nC_2} + \frac{{}^mC_2}{{}^nC_2} - \frac{{}^jC_2}{{}^nC_2}$

Source: Exercise 13.1, Q23

Given: Two cards are drawn at random from a pack of 52 cards. Find the probability that the cards are either both aces or both black cards.

Solution:

Total ways to draw 2 cards from 52: ${}^{52}C_2 = 1326$. Ways to get both aces: there are 4 aces, so ${}^4C_2 = 6$ ways, giving $P(\text{both aces}) = \frac{6}{1326}$. Ways to get both black cards: there are 26 black cards, so ${}^{26}C_2 = 325$ ways, giving $P(\text{both black}) = \frac{325}{1326}$. The overlap is “both aces and both black”, i.e. both cards are black aces; there are only 2 black aces, so ${}^2C_2 = 1$ way, giving $P(\text{overlap}) = \frac{1}{1326}$.

$$P(\text{both aces or both black}) = \frac{6}{1326} + \frac{325}{1326} - \frac{1}{1326} = \frac{330}{1326} = \frac{55}{221}$$

Final Answer

$$P = \frac{55}{221}$$

Common Mistake: Forgetting the overlap case entirely (both cards being black aces simultaneously satisfies *both* conditions), which double-counts those outcomes if not subtracted.

Quick Recall: For “both-X or both-Y” with two draws, use nC_2 throughout, and always check for a genuine overlap category before applying the addition formula.

II. Conditional Probability

Type 12

Two (or more) probabilities are given directly, including the intersection, and the question asks for a conditional probability such as $P(A|B)$ or $P(B|A)$ by plugging straight into the definition. Recognise this from simply-stated $P(A)$, $P(B)$, $P(A \cap B)$ with a request for $P(A | B)$.

Formula Used: $P(A | B) = \frac{P(A \cap B)}{P(B)}$

Source: Exercise 13.2, Q2

Given: $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{4}$. Find (i) $P(A | B)$ (ii) $P(B | A)$.

Solution:

(i) Apply the definition directly, dividing the intersection by the probability of the conditioning event B .

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/3} = \frac{1}{4} \times \frac{3}{1} = \frac{3}{4}$$

(ii) Similarly, condition on A this time by dividing by $P(A)$ instead.

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{1/4}{1/2} = \frac{1}{2}$$

Final Answer

$$P(A | B) = \frac{3}{4}, P(B | A) = \frac{1}{2}$$

Common Mistake: Dividing by the wrong probability – swapping $P(A)$ and $P(B)$ in the denominator, which mixes up $P(A|B)$ with $P(B|A)$.

Quick Recall: The event after the bar goes in the denominator: $P(A | B)$ always divides by $P(B)$.

Type 13

A conceptual/proof-style question about conditional probability itself – either a degenerate case where $P(B) = 0$ (the conditional is undefined), or a symmetry argument showing $P(A|B) = P(B|A) \Rightarrow P(A) = P(B)$. Recognise this from “what can you say

about...” or “show that...”.

Formula Used: $P(A | B) = \frac{P(A \cap B)}{P(B)}$, $P(B) \neq 0$

Source: Exercise 13.2, Q1

Given: (i) $P(A) = \frac{1}{2}$, $P(B) = 0$. What can be said about $P(A | B)$? (ii) If $P(A | B) = P(B | A)$, show that $P(A) = P(B)$.

Solution:

(i) The definition of conditional probability requires dividing by $P(B)$; since $P(B) = 0$, this division is undefined – so $P(A | B)$ simply does not exist (is not defined) in this case.

(ii) Write both conditional probabilities using the definition:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B | A) = \frac{P(A \cap B)}{P(A)}$$

Since we are told these two quantities are equal, set the right-hand sides equal to each other.

$$\frac{P(A \cap B)}{P(B)} = \frac{P(A \cap B)}{P(A)}$$

As $P(A \cap B)$ is common and nonzero (otherwise both sides are trivially 0), it cancels from both sides, leaving $P(B) = P(A)$ directly.

Final Answer

(i) $P(A | B)$ is not defined. (ii) $P(A) = P(B)$, as required.

Common Mistake: Assuming $P(A | B) = 0$ when $P(B) = 0$, rather than recognising the expression is undefined, not zero.

Quick Recall: Conditional probability needs a nonzero conditioning probability in the denominator; if $P(A|B) = P(B|A)$, the shared numerator cancels to force $P(A) = P(B)$.

Type 14

A probability question where extra information narrows the sample space (e.g. “given the sum is 6”, “given the sum is even”), and the conditional probability is found by directly counting favourable outcomes within this smaller, restricted sample space rather than by the algebraic formula. Recognise this from “given that...” combined with a small, listable sample space.

Formula Used: $P(A | B) = \frac{\text{favourable outcomes in the reduced space}}{\text{total outcomes in the reduced space}}$

Source: Exercise 13.2, Q7

Given: A die is thrown twice and the sum of the numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once?

Solution:

First restrict the sample space to only those outcomes where the sum is 6: (1, 5), (2, 4), (3, 3), (4, 2), (5, 1) – exactly 5 equally likely outcomes. Among these 5 outcomes, check which ones contain a 4 on either die: (2, 4) and (4, 2) qualify, so 2 of the 5 outcomes are favourable.

$$P(4 \text{ appears atleast once} \mid \text{sum} = 6) = \frac{2}{5}$$

Final Answer

$$P = \frac{2}{5}$$

Common Mistake: Working with the full 36-outcome sample space instead of restricting to the 5 outcomes where the sum is already known to be 6.

Quick Recall: When a condition is given, first shrink the sample space to only the outcomes satisfying that condition, then count favourably within that smaller space.

Type 15

A classic reduced-sample-space conditional probability problem, usually involving a family with two children or numbered items, asking for probabilities conditioned on partial information about birth order or a known range. Recognise this from “family has two children” or “given the number is greater than...”.

★ **Board Favorite** – The “family with two children” conditional probability problem is one of the most iconic and frequently recurring ISC/NCERT patterns in this topic.

Formula Used: $P(A \mid B) = \frac{\text{favourable outcomes in reduced space}}{\text{total outcomes in reduced space}}$

Source: Exercise 13.2, Q10

Given: A family has two children. (i) What is the probability that both children are boys, given that atleast one of them is a boy? (ii) What is the probability that both are boys if it is known that the elder child is a boy? (iii) What is the probability both are girls if the elder child is a girl?

Solution:

The full sample space (by birth order, elder first) is $\{BB, BG, GB, GG\}$, each with probability $\frac{1}{4}$.

(i) “Atleast one boy” restricts the space to $\{BB, BG, GB\}$ – 3 outcomes. Only BB has both boys.

$$P(\text{both boys} \mid \text{atleast one boy}) = \frac{1}{3}$$

(ii) “Elder is a boy” restricts the space to $\{BB, BG\}$ – 2 outcomes. Only BB has both boys.

$$P(\text{both boys} \mid \text{elder is a boy}) = \frac{1}{2}$$

(iii) “Elder is a girl” restricts the space to $\{GB, GG\}$ – 2 outcomes. Only GG has both girls.

$$P(\text{both girls} \mid \text{elder is a girl}) = \frac{1}{2}$$

Final Answer

$$(i) \frac{1}{3} \quad (ii) \frac{1}{2} \quad (iii) \frac{1}{2}$$

Common Mistake: Confusing “atleast one is a boy” (which keeps BG and GB as distinct, separate outcomes, giving denominator 3) with “a specific (elder) child is a boy” (which gives denominator 2) – these are genuinely different conditions with different answers.

Alternate Board-Style Example

Source: Exercise 13.2, Q11

Given: 12 cards numbered 1 to 12 are placed in a box and one is drawn at random. Given that the number on the card is greater than 5, find the probability that it is an odd number.

Solution:

Restrict the sample space to numbers greater than 5: $\{6, 7, 8, 9, 10, 11, 12\}$ – 7 equally likely values. Among these, the odd numbers are $\{7, 9, 11\}$ – 3 values.

$$P(\text{odd} \mid > 5) = \frac{3}{7}$$

Final Answer

$$P = \frac{3}{7}$$

Quick Recall: Draw out (or list) the reduced sample space explicitly for “given that...” problems – these are almost always easier by direct counting than by the algebraic formula.

Type 16

Two events are defined on a card draw using overlapping descriptions (e.g. “king or queen” and “queen or ace”), and the conditional probability of one given the other is required. Recognise this from two events defined by rank categories that share a common rank.

Formula Used: $P(E | F) = \frac{P(E \cap F)}{P(F)}$

Source: Exercise 13.2, Q13

Given: One card is drawn from a well-shuffled pack of 52 cards. E = “the card drawn is a king or a queen”, F = “the card drawn is a queen or an ace”. Find $P(E | F)$.

Solution:

Count F first: queens and aces together number $4 + 4 = 8$ cards, so $P(F) = \frac{8}{52}$. Now find $E \cap F$ – cards that are simultaneously (king or queen) *and* (queen or ace): only the 4 queens satisfy both descriptions, so $P(E \cap F) = \frac{4}{52}$.

$$P(E | F) = \frac{P(E \cap F)}{P(F)} = \frac{4/52}{8/52} = \frac{4}{8} = \frac{1}{2}$$

Final Answer

$$P(E | F) = \frac{1}{2}$$

Common Mistake: Miscounting the intersection – forgetting that only the queens (the rank common to both descriptions) belong to $E \cap F$, not kings or aces.

Quick Recall: When two events are described by overlapping rank lists, find the shared rank(s) directly to get $E \cap F$, then apply the conditional formula.

Type 17

A conditional probability problem involving two dice where the given condition is an inequality (e.g. “sum more than 9”), requiring the full outcome list satisfying the inequality to be written out before counting. Recognise this from “I roll two dice and get a sum more than...”.

Formula Used: $P(A | B) = \frac{\text{favourable outcomes in reduced space}}{\text{total outcomes in reduced space}}$

Source: Exercise 13.2, Q14

Given: I roll two dice and get a sum more than 9. What is the probability that the number on the first die is even?

Solution:

List every outcome with sum > 9 , i.e. sum ≥ 10 : $(4, 6), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)$ – exactly 6 outcomes. Among these, check which have an even first die: $(4, 6), (6, 4), (6, 5), (6, 6)$ qualify – 4 outcomes.

$$P(\text{first die even} \mid \text{sum} > 9) = \frac{4}{6} = \frac{2}{3}$$

Final Answer

$$P = \frac{2}{3}$$

Common Mistake: Missing outcomes when listing the reduced sample space (e.g. forgetting $(5, 6)$ or double-listing $(6, 6)$), which throws off both the numerator and denominator.

Quick Recall: For dice “sum more than k ” conditions, systematically list pairs starting from the highest first-die value downward to avoid missing any.

Type 18

A word problem gives percentages of a population satisfying two overlapping properties (e.g. reading two newspapers), and asks for conditional probabilities between the two properties. Recognise this from “X% read/passing A, Y% read/passing B, Z% read/passing both”.

Formula Used: $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$

Source: Exercise 13.2, Q16

Given: 60% students read Hindi newspaper, 40% read Tamil newspaper, and 20% read both. Find the probability that a randomly selected student reads (i) Tamil given Hindi (ii) Hindi given Tamil (iii) neither.

Solution:

Convert percentages to probabilities: $P(H) = 0.6$, $P(T) = 0.4$, $P(H \cap T) = 0.2$. (i)

$$P(T \mid H) = \frac{P(H \cap T)}{P(H)} = \frac{0.2}{0.6} = \frac{1}{3}$$

(ii)

$$P(H \mid T) = \frac{P(H \cap T)}{P(T)} = \frac{0.2}{0.4} = \frac{1}{2}$$

(iii) Neither is the complement of the union; first find the union.

$$P(H \cup T) = 0.6 + 0.4 - 0.2 = 0.8 \implies P(\text{neither}) = 1 - 0.8 = 0.2 = \frac{1}{5}$$

Final Answer

(i) $\frac{1}{3}$ (ii) $\frac{1}{2}$ (iii) $\frac{1}{5}$

Common Mistake: Dividing by the wrong percentage – using $P(T)$ in the denominator when finding $P(T|H)$ instead of $P(H)$.

Quick Recall: Convert percentages to decimals immediately, then treat the problem as an ordinary conditional-probability-plus-addition-theorem question.

Type 19

A conditional probability question layered on top of combinatorics – a committee or group is selected, and the condition restricts to a sub-scenario (e.g. “atleast one girl”) while the target event asks for a more specific composition (e.g. “exactly 2 boys”). Recognise this from committee-selection language combined with “given that...”.

Formula Used: $P(E | F) = \frac{P(E \cap F)}{P(F)}$

Formula Used: ${}^nC_r = \frac{n!}{r!(n-r)!}$

Source: Exercise 13.2, Q17

Given: A committee of 4 students is selected from 7 boys and 4 girls. Find the probability that exactly 2 boys are in the committee, given that atleast one girl must be there.

Solution:

Total ways to pick any 4 from 11 students: ${}^{11}C_4 = 330$. F = “atleast one girl”; its complement is “no girls at all” (all 4 boys): ${}^7C_4 = 35$ ways.

$$P(F) = 1 - \frac{35}{330} = \frac{295}{330}$$

$E \cap F$ = “exactly 2 boys and atleast one girl”; since 2 boys out of 4 automatically leaves 2 girls, this is simply “exactly 2 boys and 2 girls”: ${}^7C_2 \times {}^4C_2 = 21 \times 6 = 126$ ways.

$$P(E \cap F) = \frac{126}{330}$$

$$P(E | F) = \frac{126/330}{295/330} = \frac{126}{295}$$

Final Answer

$$P = \frac{126}{295}$$

Common Mistake: Computing $P(E)$ directly (unconditioned) and forgetting to divide by $P(F)$, or miscounting the complement of “atleast one girl” as something other than “all boys”.

Quick Recall: Translate “atleast one girl” into its easier complement (“no girls”), and note that “exactly 2 boys” out of a committee of 4 forces the remaining 2 seats to girls automatically.

Type 20

A logic-based conditional probability question about the arrangement or positioning of people or objects, where the condition fixes one element’s position and the target asks about another element’s position. Recognise this from spatial phrases like “standing at the left corner”, “standing at the central position”.

Formula Used: $P(A | B) = \frac{\text{favourable arrangements in reduced space}}{\text{total arrangements in reduced space}}$

Source: Exercise 13.2, Q18

Given: Three friends A, B, C got their photograph clicked. Find the probability that B is standing at the central position, given that A is standing at the left corner.

Solution:

Fix A at the left corner as given; the remaining two positions (centre, right) can be filled by B and C in $2! = 2$ ways: (A, B, C) and (A, C, B) . Among these 2 equally likely arrangements, B is in the centre in exactly 1 of them, namely (A, B, C) .

$$P(B \text{ central} | A \text{ left}) = \frac{1}{2}$$

Final Answer

$$P = \frac{1}{2}$$

Common Mistake: Forgetting to fix A’s position first and instead working with all $3! = 6$ arrangements of three people, which ignores the given condition entirely.

Quick Recall: When a condition fixes one person’s position, only permute the remaining people among the remaining positions – don’t recount the whole arrangement space.

III. Multiplication Theorem & Independent Events

Type 21

A conceptual fill-in-the-blank question testing the core distinction between mutually exclusive and independent events, and how their intersection probability behaves under replacement/without replacement draws. Recognise this from “fill in the blanks” covering both terms together.

Formula Used: Mutually exclusive: $P(A \cap B) = 0$. Independent: $P(A \cap B) = P(A) \cdot P(B)$

Source: Exercise 13.3, Q1

Given: (i) If events A and B are mutually exclusive, $P(A \cap B)$ is ____; if independent, $P(A \cap B)$ is _____. (ii) If $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$, is the statement “ A and B are independent” true, and what is $P(A \cup B)$?

Solution:

(i) By definition, mutually exclusive events cannot occur together, so $P(A \cap B) = 0$; independent events satisfy the product rule, so $P(A \cap B) = P(A) \cdot P(B)$.

(ii) Test independence by checking whether the product rule holds: $P(A) \times P(B) = 0.5 \times 0.4 = 0.2$, which exactly matches the given $P(A \cap B) = 0.2$, so the events *are* independent. Then use the addition formula for the union.

$$P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$$

Final Answer

(i) 0; $P(A)P(B)$ (ii) True; $P(A \cup B) = 0.7$

Common Mistake: Confusing the two conditions – assuming mutually exclusive events are automatically independent (they are not, unless one has probability 0).

Quick Recall: Mutually exclusive $\Rightarrow$ intersection is 0; independent $\Rightarrow$ intersection is the product – these are two different, usually incompatible, properties.

Type 22

A classical probability question on one, two, or three dice thrown together (not necessarily involving independence formulas explicitly), where the answer comes from directly counting favourable outcomes among all equally likely outcomes. Recognise this from “a pair of dice is rolled” or “three dice are thrown” with a specific numeric

target.

Formula Used: $P(\text{event}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$

Source: Exercise 13.3, Q2

Given: What is the probability of obtaining an even prime number on each die when a pair of dice is rolled?

Solution:

Among the numbers 1 to 6, the only even prime number is 2 (the sole even prime in existence). So we need both dice to individually show 2: there is exactly 1 favourable outcome, (2, 2), out of 36 total equally likely outcomes.

$$P(\text{both dice show 2}) = \frac{1}{36}$$

Final Answer

$$P = \frac{1}{36}$$

Common Mistake: Misidentifying “even prime” as any even number, forgetting that 2 is the unique number that is both even and prime.

Quick Recall: “Even prime” always means the single number 2 – don’t confuse it with “even number” in general.

Type 23

Two probabilities and their intersection (or union) are given, and the question asks you to *test* whether the events are independent by checking if the product rule $P(A)P(B) = P(A \cap B)$ actually holds. Recognise this from “state whether A and B are independent” or “are E and F independent?”.

Formula Used: Independent $\iff P(A \cap B) = P(A) \cdot P(B)$

Source: Exercise 13.3, Q3(ii)

Given: Let E and F be events with $P(E) = \frac{3}{5}$, $P(F) = \frac{3}{10}$, $P(E \cap F) = \frac{1}{5}$. Are E and F independent?

Solution:

Compute the product $P(E) \times P(F)$ and compare it directly to the given $P(E \cap F)$.

$$P(E) \times P(F) = \frac{3}{5} \times \frac{3}{10} = \frac{9}{50} = 0.18$$

The given intersection is $P(E \cap F) = \frac{1}{5} = 0.20$. Since $0.18 \neq 0.20$, the product rule fails, so the events do *not* satisfy the independence condition.

Final Answer

E and F are **not independent**.

Common Mistake: Rounding too early or comparing fractions with different denominators without converting to a common form, leading to a false “equal” conclusion.

Quick Recall: To test independence, always compute $P(A) \times P(B)$ separately and compare exactly to the given $P(A \cap B)$ – never assume.

Type 24

Two events are stated to be independent, and their individual probabilities are given; the question asks for a derived probability such as the union, using the independence product rule inside the addition formula. Recognise this from “if A and B are independent... find $P(A \cup B)$ ”.

Formula Used: $P(A \cap B) = P(A) \cdot P(B)$ (independent events)

Formula Used: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Source: Exercise 13.3, Q4(iii)

Given: Events A and B are independent such that $P(A) = \frac{3}{5}$, $P(B) = \frac{2}{3}$. Find $P(A \cup B)$.

Solution:

Since the events are independent, first compute the intersection as a simple product.

$$P(A \cap B) = \frac{3}{5} \times \frac{2}{3} = \frac{6}{15} = \frac{2}{5}$$

Substitute into the addition formula, using a common denominator of 15.

$$P(A \cup B) = \frac{3}{5} + \frac{2}{3} - \frac{2}{5} = \frac{9}{15} + \frac{10}{15} - \frac{6}{15} = \frac{13}{15}$$

Final Answer

$$P(A \cup B) = \frac{13}{15}$$

Common Mistake: Forgetting that independence gives $P(A \cap B) = P(A)P(B)$ specifically – and instead assuming the events are mutually exclusive (which would wrongly skip the subtraction).

Quick Recall: “Independent” is your permission to multiply for the intersection – always compute that product first before touching the addition formula.

Type 25

Two events are independent, but one probability is missing and must be found algebraically from a given union or complement value. Recognise this from independence stated alongside an equation that needs solving for an unknown probability.

Formula Used: $P(A \cup B) = P(A) + P(B) - P(A)P(B)$ (independent events)

Source: Exercise 13.3, Q4(iv)

Given: A and B are independent events such that $P(\text{not } B) = 0.65$ and $P(A \cup B) = 0.85$. Find $P(A)$.

Solution:

First convert the complement to $P(B)$ directly.

$$P(B) = 1 - 0.65 = 0.35$$

Substitute into the independent-events union formula, keeping $P(A)$ as the unknown.

$$0.85 = P(A) + 0.35 - 0.35 P(A)$$

Group the $P(A)$ terms together on one side to solve.

$$0.85 - 0.35 = P(A)(1 - 0.35) \implies 0.5 = 0.65 P(A)$$

$$P(A) = \frac{0.5}{0.65} = \frac{50}{65} = \frac{10}{13}$$

Final Answer

$$P(A) = \frac{10}{13}$$

Common Mistake: Treating $P(A) - 0.35P(A)$ as $0.65P(A)$ incorrectly by mixing up the coefficient, or forgetting to factor $P(A)$ out before dividing.

Quick Recall: For independent events, the union formula becomes $P(A) + P(B) - P(A)P(B)$ – factor out the unknown probability carefully when solving.

Type 26

Multiple independent trials or events are given (often 3 or more), and the question asks for the probability that *at least one* succeeds. Recognise this from a list of individual success probabilities across trials/shots/products combined with “at least one”.

★ **Board Favorite** – “At least one success across independent trials” is a very high-yield ISC pattern – it appeared as an ISC 2024 board question and recurs almost every year in some form.

Formula Used: $P(\text{at least one success}) = 1 - P(\text{all fail}) = 1 - \prod(1 - p_i)$

Source: Exercise 13.3, Q4(vii)

Given: Teena’s probabilities of hitting the target in her 1st, 2nd, 3rd and 4th shots are 0.4, 0.3, 0.2, 0.1 respectively. Find the probability of at least one shot hitting the target.

Solution:

It’s far easier to find the complement – the probability that *every* shot misses – since “at least one hit” is hard to count directly but “all miss” is a straightforward product (shots are independent). Probabilities of missing each shot: 0.6, 0.7, 0.8, 0.9.

$$P(\text{all miss}) = 0.6 \times 0.7 \times 0.8 \times 0.9 = 0.3024$$

Now subtract from 1 to get at least one hit.

$$P(\text{at least one hit}) = 1 - 0.3024 = 0.6976 \approx 0.697$$

Final Answer

$$P(\text{at least one hit}) \approx 0.697 \text{ (i.e. } \frac{436}{625})$$

Common Mistake: Trying to directly add the four individual hit-probabilities (which overcounts overlapping successes) instead of using the far simpler complement approach.

Alternate Board-Style Example

Source: Exercise 13.3, Q42

Given: A policeman fires four bullets at a dacoit. The probability that the dacoit is killed by one bullet is 0.6. What is the probability that the dacoit is still alive?

Solution:

“Still alive” means every one of the 4 independent shots missed.

$$P(\text{miss one shot}) = 1 - 0.6 = 0.4$$

$$P(\text{dacoit alive}) = (0.4)^4 = 0.0256 = \frac{16}{625}$$

Final Answer

$$P = \frac{16}{625}$$

Quick Recall: Whenever you see “atleast one” with several independent trials, immediately switch to the complement: $1 - P(\text{all fail})$.

Type 27

A real-world scenario gives probabilities for two events and their intersection, and asks the student to classify the relationship as independent, mutually exclusive, both, or neither – often as an MCQ or by testing several event-pairs from the same experiment. Recognise this from “which one of the following is true about the events” or listing pairs E_1, E_2 to classify.

★ **Board Favorite** – Classify-the-relationship MCQs (independent vs mutually exclusive vs neither) are a recurring ISC Specimen Paper pattern testing conceptual understanding rather than computation.

Formula Used: Mutually exclusive $\iff P(A \cap B) = 0$. Independent $\iff P(A \cap B) = P(A)P(B)$

Source: Exercise 13.3, Q7

Given: Bhuvan prepares chapati and curry. A = chapati is prepared well, B = curry is cooked well. $P(A) = 0.40$, $P(B) = 0.30$, $P(A \text{ and } B) = 0.20$. Which is true: (a) mutually exclusive not independent (b) independent not mutually exclusive (c) both (d) neither?

Solution:

Test independence: $P(A) \times P(B) = 0.40 \times 0.30 = 0.12$, but the given $P(A \cap B) = 0.20$. Since $0.12 \neq 0.20$, the events are **not independent**. Test mutually exclusive: $P(A \cap B) = 0.20 \neq 0$, so the events *do* overlap, meaning they are **not mutually exclusive** either. Since neither condition holds, the correct choice is (d).

Final Answer

(d) They are neither mutually exclusive nor independent.

Common Mistake: Assuming that because two events *can* occur together (nonzero intersection), they must automatically be independent – overlap and independence are separate, unrelated tests.

Alternate Board-Style Example

Source: Exercise 13.3, Q8(i)

Given: A coin is tossed thrice; all 8 outcomes are equally likely. E_1 : the first throw results in head. E_2 : the last throw results in tail. Are E_1 and E_2 independent? Mutually exclusive?

Solution:

Sample space: $\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$, each with probability $\frac{1}{8}$. E_1 (first = H): $\{HHH, HHT, HTH, HTT\}$, so $P(E_1) = \frac{4}{8} = \frac{1}{2}$. E_2 (last = T): $\{HHT, HTT, THT, TTT\}$, so $P(E_2) = \frac{4}{8} = \frac{1}{2}$. $E_1 \cap E_2$ (first H and last T): $\{HHT, HTT\}$, so $P(E_1 \cap E_2) = \frac{2}{8} = \frac{1}{4}$. Check the product: $P(E_1) \times P(E_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$, which matches – so E_1, E_2 are **independent**. Since $E_1 \cap E_2 \neq \phi$, they are **not mutually exclusive**.

Final Answer

Independent, but not mutually exclusive.

Quick Recall: Always run both tests separately – compute $P(A \cap B)$ once, then compare it to 0 (exclusive test) and to $P(A)P(B)$ (independence test).

Type 28

A group of students split into categories (boys/girls) forms a team via combinations, and the probability of a specific mixed composition is required. Recognise this from “find the probability that k boys and m girls are selected” using nC_r .

Formula Used: $P(\text{specific composition}) = \frac{{}^aC_k \times {}^bC_m}{{}^nC_{k+m}}$

Source: Exercise 13.3, Q9(i)

Given: Out of 8 outstanding students (3 boys, 5 girls), a team of 4 is selected. Find the probability that 2 boys and 2 girls are selected.

Solution:

Total ways to choose any 4 students from 8: ${}^8C_4 = 70$. Favourable ways: choose 2 boys from 3 and 2 girls from 5, and multiply since both choices happen together.

$${}^3C_2 \times {}^5C_2 = 3 \times 10 = 30$$

$$P(2 \text{ boys, } 2 \text{ girls}) = \frac{30}{70} = \frac{3}{7}$$

Final Answer

$$P = \frac{3}{7}$$

Common Mistake: Adding ${}^3C_2 + {}^5C_2$ instead of multiplying – forgetting that choosing boys and girls are simultaneous, independent sub-choices that combine multiplicatively.

Quick Recall: For “ k of type A and m of type B”, multiply the two separate combination counts, then divide by the overall ${}^nC_{k+m}$.

Type 29

Two items are drawn one by one *without* replacement, and a specific mixed outcome (e.g. one red then one black, or in either order) is required, computed via sequential conditional probabilities rather than combinations. Recognise this from “drawn one by one without replacement” with a target combination.

Formula Used: $P(\text{first} = X, \text{second} = Y) = P(X) \times P(Y \mid X \text{ drawn})$

Source: Exercise 13.3, Q9(ii)

Given: Two cards are drawn at random, one by one without replacement, from a well-shuffled pack of 52 cards. Find the probability that one card is red and the other is black.

Solution:

This can happen two ways – red then black, or black then red – so compute each sequential probability and add them.

$$P(\text{red, then black}) = \frac{26}{52} \times \frac{26}{51}$$

The second fraction uses 51 (one card already removed) and 26 (black count unaffected by removing a red card).

$$P(\text{black, then red}) = \frac{26}{52} \times \frac{26}{51}$$

By symmetry both sequential probabilities are identical here, so add them.

$$P(\text{one red, one black}) = 2 \times \frac{26}{52} \times \frac{26}{51} = \frac{26}{51}$$

Final Answer

$$P = \frac{26}{51}$$

Common Mistake: Forgetting to update the denominator to 51 for the second draw (since one card has already been removed without replacement).

Quick Recall: Without replacement, every draw after the first uses one fewer card in both the count and total – and remember to account for both possible orders when the target doesn't specify one.

Type 30

Two items are drawn (or two experiments repeated) *with* replacement, so the draws are genuinely independent of each other, and a specific combined outcome (e.g. both even numbers) is required by simply multiplying identical individual probabilities. Recognise this from “with replacement” language.

Formula Used: $P(\text{two independent identical draws}) = [P(\text{single draw})]^2$

Source: Exercise 13.3, Q10

Given: A bag contains 19 tickets, numbered 1 to 19. Two tickets are drawn in succession *with* replacement. Find the probability that both tickets drawn are even numbered.

Solution:

Even numbers from 1 to 19: 2, 4, 6, ..., 18 – that's 9 even numbers out of 19 total, so $P(\text{even on one draw}) = \frac{9}{19}$. Since the ticket is replaced before the second draw, the two draws are independent and identical in probability.

$$P(\text{both even}) = \frac{9}{19} \times \frac{9}{19} = \frac{81}{361}$$

Final Answer

$$P = \frac{81}{361}$$

Common Mistake: Using 18 as the denominator for the second draw out of habit (as if without replacement), even though the ticket was explicitly replaced.

Quick Recall: “With replacement” always keeps the denominator and composition unchanged for every draw – just multiply the same single-draw probability repeatedly.

Type 31

Several independent events (often 3 people, machines, or products) each have their own probability of success, and the question asks for the probability that *at least one* succeeds – solved via the complement of “none succeed”. Recognise this from three or more distinct success probabilities combined with “at least one is successful”.

Formula Used: $P(\text{atleast one succeeds}) = 1 - \prod_i (1 - p_i)$

Source: Exercise 13.3, Q11

Given: A company estimates the probabilities of success for three products as $\frac{1}{3}, \frac{2}{5}, \frac{2}{3}$ respectively. Assuming independence, find the probability that atleast one product is successful.

Solution:

Find the probability that all three products fail, using the complements of the given success probabilities.

$$P(\text{all fail}) = \left(1 - \frac{1}{3}\right) \left(1 - \frac{2}{5}\right) \left(1 - \frac{2}{3}\right) = \frac{2}{3} \times \frac{3}{5} \times \frac{1}{3}$$

Multiply the fractions carefully, cancelling common factors where possible.

$$P(\text{all fail}) = \frac{2 \times 3 \times 1}{3 \times 5 \times 3} = \frac{6}{45} = \frac{2}{15}$$

Subtract from 1 to get the atleast-one-success probability.

$$P(\text{atleast one success}) = 1 - \frac{2}{15} = \frac{13}{15}$$

Final Answer

$$P = \frac{13}{15}$$

Common Mistake: Trying to add the three success probabilities directly $(\frac{1}{3} + \frac{2}{5} + \frac{2}{3})$, which is not a valid probability computation and can even exceed 1.

Quick Recall: For “atleast one of several independent things succeeds”, always go through the complement – multiply the individual failure probabilities and subtract from 1.

Type 32

A fixed number of balls/discs are drawn one by one *without* replacement, and the target event is that all draws share the same property (all white, no ball white, all blue, etc.), computed as a chain of shrinking conditional probabilities. Recognise this from “drawn one by one without replacement, find the probability that all are...” or “no ball is...”.

★ **Board Favorite** – Sequential without-replacement “all same category” problems are extremely common across ISC papers and test careful denominator-shrinking, a skill boards specifically reward.

Formula Used: $P(\text{all } k \text{ draws same category}) = \frac{a}{n} \times \frac{a-1}{n-1} \times \cdots \times \frac{a-k+1}{n-k+1}$

Source: Exercise 13.3, Q12(i)

Given: A bag contains 4 white, 3 red and 2 blue balls. Four balls are drawn one by one without replacement. Find the probability that all balls are white.

Solution:

There are only 4 white balls out of 9 total, and we need all 4 draws to be white – so every draw must succeed, shrinking both the white count and the total each time.

$$P(\text{all white}) = \frac{4}{9} \times \frac{3}{8} \times \frac{2}{7} \times \frac{1}{6}$$

Multiply step by step, cancelling as you go.

$$= \frac{4 \times 3 \times 2 \times 1}{9 \times 8 \times 7 \times 6} = \frac{24}{3024} = \frac{1}{126}$$

Final Answer

$$P(\text{all white}) = \frac{1}{126}$$

Common Mistake: Forgetting to shrink *both* the numerator (remaining white balls) and denominator (remaining total balls) at every step – shrinking only the denominator overstates the probability.

Alternate Board-Style Example

Source: Exercise 13.3, Q13(i)

Given: Using the same bag (4 white, 3 red, 2 blue), four balls are drawn one by one without replacement. Find the probability that no ball is white.

Solution:

“No ball white” means all 4 draws come from the 5 non-white balls (3 red + 2 blue), again shrinking both counts each draw.

$$P(\text{no white}) = \frac{5}{9} \times \frac{4}{8} \times \frac{3}{7} \times \frac{2}{6} = \frac{120}{3024} = \frac{5}{126}$$

Final Answer

$$P(\text{no white}) = \frac{5}{126}$$

Quick Recall: Without replacement, chain the fractions with both numerator and denominator decreasing by 1 at every step – multiply all the way through.

Type 33

The same style of “all draws from one category” question, but explicitly *with* replacement, so every draw uses the identical, unchanging probability – this becomes a simple power rather than a shrinking chain. Recognise this from “drawn one after the other with replacement”.

Formula Used: $P(\text{all } k \text{ draws same category, with replacement}) = \left(\frac{a}{n}\right)^k$

Source: Exercise 13.3, Q12(ii)

Given: The same bag (4 white, 3 red, 2 blue out of 9) has four balls drawn one by one *with* replacement. Find the probability that all balls are white.

Solution:

With replacement, the bag composition is restored after every draw, so each draw independently has the same probability of white, $\frac{4}{9}$.

$$P(\text{all white}) = \left(\frac{4}{9}\right)^4$$

This is simply the single-draw probability raised to the power of the number of draws – no shrinking needed since nothing changes between draws.

Final Answer

$$P(\text{all white}) = \left(\frac{4}{9}\right)^4$$

Common Mistake: Accidentally shrinking the fractions as in the without-replacement case, forgetting that replacement keeps every draw identical.

Quick Recall: “With replacement” turns a shrinking chain into a clean power: just raise the single-draw probability to the number of draws.

Type 34

Two balls (or cards) are drawn together (using combinations), and the target is a union of two possible “both-same-category” outcomes (e.g. both red or both black). Recognise this from “both X or both Y” phrased with balls/discs rather than cards (compare Type 11).

Formula Used: $P(\text{both A or both B}) = \frac{{}^aC_2 + {}^bC_2}{{}^nC_2}$

Source: Exercise 13.3, Q14(i)

Given: A bag contains 5 red, 6 white and 7 black balls. Two balls are drawn at random. Find the probability that both balls are red or both are black.

Solution:

Total ways to draw 2 balls from 18: ${}^{18}C_2 = 153$. Since “both red” and “both black” cannot happen at the same time (a pair can’t be both colours simultaneously), these are genuinely mutually exclusive outcomes – no overlap to subtract.

$${}^5C_2 = 10, \quad {}^7C_2 = 21$$

$$P(\text{both red or both black}) = \frac{10 + 21}{153} = \frac{31}{153}$$

Final Answer

$$P = \frac{31}{153}$$

Common Mistake: Subtracting an overlap term out of habit (copying the Type 11 pattern) even though “both red” and “both black” are genuinely mutually exclusive here – there is nothing to subtract.

Quick Recall: “Both X or both Y” with two *different* colours from the same pair is always mutually exclusive – just add the two combination counts.

Type 35

Balls or items are drawn one by one without replacement, and the question specifies an *exact order* (e.g. white first, then red, then red again), so the sequential probabilities are multiplied in that specific fixed order rather than considering all arrangements. Recognise this from “drawing a white ball followed by two red balls”.

Formula Used: $P(\text{specific fixed sequence}) = P(1^{st}) \times P(2^{nd} | 1^{st}) \times P(3^{rd} | 1^{st}, 2^{nd})$

Source: Exercise 13.3, Q16

Given: A bag contains 1 white, 5 red and 4 black balls. If three balls are drawn one by one without replacement, find the probability of drawing a white ball followed by two red balls.

Solution:

The order is fixed: white first, so use the original counts; then red, from the reduced bag after the white ball is removed; then red again, from the further-reduced bag.

$$P(\text{white}) = \frac{1}{10}$$

After removing the white ball, 9 balls remain (5 red, 4 black):

$$P(\text{red} | \text{white already drawn}) = \frac{5}{9}$$

After removing one red ball too, 8 balls remain (4 red, 4 black):

$$P(\text{red} \mid \text{white, red already drawn}) = \frac{4}{8}$$

Multiply the three sequential probabilities together.

$$P(W, R, R) = \frac{1}{10} \times \frac{5}{9} \times \frac{4}{8} = \frac{20}{720} = \frac{1}{36}$$

Final Answer

$$P = \frac{1}{36}$$

Common Mistake: Forgetting that a fixed order means only *one* sequence counts – unlike Type 29, there is no need to consider multiple orderings and add them.

Quick Recall: When a specific order is stated (“followed by”), multiply straight through in that exact order with shrinking denominators – do not multiply by any extra arrangement factor.

Type 36

Two items are drawn together (via combinations) from a numbered set, and the target event depends on a number-theoretic property such as being prime, or one satisfying one property while the other satisfies a different property. Recognise this from “both have prime numbers” or “one is prime, the other is a multiple of...”.

Formula Used: $P(\text{event}) = \frac{\text{favourable } C \text{ count}}{{}^nC_2}$

Source: Exercise 13.3, Q17(i)

Given: From a bag containing 20 tickets numbered 1 to 20, two tickets are drawn at random. Find the probability that both the tickets have prime numbers on them.

Solution:

List the primes between 1 and 20: 2, 3, 5, 7, 11, 13, 17, 19 – exactly 8 primes. Total ways to draw 2 tickets from 20: ${}^{20}C_2 = 190$. Favourable ways: both drawn from the 8 primes: ${}^8C_2 = 28$.

$$P(\text{both prime}) = \frac{28}{190} = \frac{14}{95}$$

Final Answer

$$P = \frac{14}{95}$$

Common Mistake: Forgetting to list out the actual primes carefully (a common slip is missing 2, the only even prime, or mistakenly including 1, which is not prime).

Quick Recall: Always write out the qualifying numbers explicitly first (primes, multiples, etc.) before applying combinations – guessing the count leads straight to an error.

Type 37

Two independent sources (bags, urns, boxes) are each sampled once, and the question asks for the probability that both draws share the same property (e.g. same colour), computed by summing the products of matching-category probabilities across the two sources. Recognise this from “one ball is drawn from each of two boxes... same colour”.

Formula Used: $P(\text{same category from two independent sources}) = \sum_i P(\text{category}_i \text{ from source 1}) \times P(\text{category}_i \text{ from source 2})$

Source: Exercise 13.3, Q22(i)

Given: Box B_1 contains 1 white and 3 red balls. Box B_2 contains 2 white and 3 red balls. One ball is drawn at random from each box. Find the probability that the two balls drawn are of the same colour.

Solution:

“Same colour” can happen in two independent ways – both white, or both red – so compute each and add, since drawing from the two boxes are independent events.

$$P(\text{both white}) = \frac{1}{4} \times \frac{2}{5} = \frac{2}{20}$$

$$P(\text{both red}) = \frac{3}{4} \times \frac{3}{5} = \frac{9}{20}$$

$$P(\text{same colour}) = \frac{2}{20} + \frac{9}{20} = \frac{11}{20}$$

Final Answer

$$P = \frac{11}{20}$$

Common Mistake: Multiplying probabilities from the wrong box (mixing up B_1 's and B_2 's totals) instead of keeping each box's own denominator consistent.

Quick Recall: When drawing once from each of two separate sources, multiply straight across for a matching category, then add across all the matching categories.

Type 38

A group split into two categories (boys/girls) has a fixed number of members chosen, and the target is a union of two specific compositions (e.g. 3 boys+1 girl OR 3 girls+1 boy). Recognise this from “either...and...or...and...” phrasing describing two distinct,

mutually exclusive team compositions.

Formula Used: $P(\text{composition A or composition B}) = \frac{C\text{-count(A)} + C\text{-count(B)}}{{}^nC_r}$

Source: Exercise 13.3, Q24

Given: In a group of students, there are 3 boys and 3 girls. Four students are selected at random. Find the probability that either 3 boys and 1 girl, or 3 girls and 1 boy, are selected.

Solution:

Total ways to choose 4 from 6 students: ${}^6C_4 = 15$. These two compositions (3 boys+1 girl, and 3 girls+1 boy) cannot happen simultaneously in the same selection, so they are mutually exclusive – just add their counts.

$${}^3C_3 \times {}^3C_1 = 1 \times 3 = 3 \quad \text{and} \quad {}^3C_3 \times {}^3C_1 = 1 \times 3 = 3$$

$$P = \frac{3+3}{15} = \frac{6}{15} = \frac{2}{5}$$

Final Answer

$$P = \frac{2}{5}$$

Common Mistake: Miscounting ${}^3C_3 = 1$ as something other than 1 – students sometimes forget that choosing all 3 available boys leaves exactly one way to do so.

Quick Recall: Two distinct fixed compositions of the same total group size are automatically mutually exclusive – compute each combination product separately, then add.

Type 39

Several independent people/agents each attempt a task with their own success probability, and the question asks for the probability of an *exact* number of successes (not just at least one), requiring you to enumerate which specific combination of successes/failures gives that exact count. Recognise this from “exactly two persons hit the target” or “exactly one of them solves the problem”.

Formula Used: $P(\text{exactly } k \text{ successes}) = \sum(\text{all ways to choose which } k \text{ succeed, others fail})$

Source: Exercise 13.3, Q33(i)

Given: Three persons A, B, C shoot at a target with probabilities $\frac{5}{6}, \frac{4}{5}, \frac{3}{4}$ respectively. Find the probability that exactly two persons hit the target.

Solution:

“Exactly two hit” can happen in three distinct, mutually exclusive ways: (A,B hit, C misses), (A,C hit, B misses), or (B,C hit, A misses). Compute each and add.

$$P(A, B \text{ hit, } C \text{ miss}) = \frac{5}{6} \times \frac{4}{5} \times \frac{1}{4} = \frac{20}{120}$$

$$P(A, C \text{ hit, } B \text{ miss}) = \frac{5}{6} \times \frac{1}{5} \times \frac{3}{4} = \frac{15}{120}$$

$$P(B, C \text{ hit, } A \text{ miss}) = \frac{1}{6} \times \frac{4}{5} \times \frac{3}{4} = \frac{12}{120}$$

Add the three mutually exclusive cases together.

$$P(\text{exactly two hit}) = \frac{20 + 15 + 12}{120} = \frac{47}{120}$$

Final Answer

$$P = \frac{47}{120}$$

Common Mistake: Missing one of the three combinations, or using the hit-probability instead of the miss-probability for the person who is supposed to fail in each case.

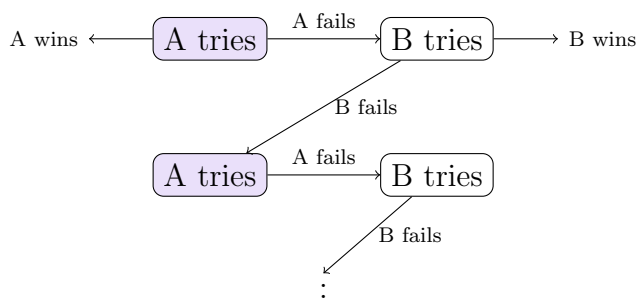
Quick Recall: “Exactly k successes” means listing every way to pick which agents succeed and which fail, computing each product, and adding all the cases together.

Type 40

Two players take turns performing an experiment (tossing a coin, rolling dice, drawing a card) repeatedly, and the game continues until someone achieves a target outcome; the probability that a specific player wins is found by summing an infinite geometric series over the rounds in which that player could win. Recognise this from “A and B throw/toss alternately till one of them gets...”.

★ **Board Favorite** – Games-played-alternately problems (infinite geometric series of turns) are a distinctive, frequently tested ISC pattern precisely because they combine probability with the geometric series formula from a different chapter.

Formula Used: $P(\text{first player wins}) = \frac{p}{1 - qr}$ where $q = 1 - p$ (if only one event type per turn)



Turn structure: A and B alternate attempts; the game only continues into a further round if everyone in the current round fails.

Source: Exercise 13.3, Q43

Given: A and B toss a coin alternately till one of them gets a head and wins the game. A starts first. Find the probability that B wins the game.

Solution:

B can only win on the 2nd, 4th, 6th, ... toss overall – meaning A must fail every single time before B's winning toss, and every earlier B-toss must also have failed. The probability B wins on exactly the 2nd toss (A fails, B succeeds): $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. The probability B wins on the 4th toss overall (A fails, B fails, A fails, B succeeds): $\left(\frac{1}{2}\right)^3 \times \frac{1}{2} = \frac{1}{16}$. This pattern continues with common ratio $\frac{1}{4}$ each full round, forming an infinite geometric series.

$$P(B \text{ wins}) = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \left(\frac{1}{4}\right)^2 + \cdots = \frac{1/4}{1 - 1/4} = \frac{1/4}{3/4} = \frac{1}{3}$$

Final Answer

$$P(B \text{ wins}) = \frac{1}{3}$$

Common Mistake: Forgetting this is an *infinite* series (the game could in principle continue forever) and instead only computing the probability of B winning on the very first attempt available to them.

Alternate Board-Style Example

Source: Exercise 13.3, Q47

Given: A and B cut a pack of 52 playing cards alternately (with reshuffling after each cut) until one of them cuts a heart and wins. A starts. Find their respective probabilities of winning.

Solution:

Each cut, $P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$, and $P(\text{not heart}) = \frac{3}{4}$. A wins on cut 1, 3, 5, ... with

probability $\frac{1}{4} + \left(\frac{3}{4}\right)^2 \frac{1}{4} + \dots = \frac{1/4}{1 - 9/16} = \frac{1/4}{7/16} = \frac{4}{7}$. Since exactly one of A or B must eventually win, $P(B \text{ wins}) = 1 - \frac{4}{7} = \frac{3}{7}$.

Final Answer

$$P(A) = \frac{4}{7}, P(B) = \frac{3}{7}$$

Quick Recall: Games played alternately always reduce to an infinite geometric series – identify the “both fail” round probability as the common ratio, then sum using $\frac{\text{first term}}{1 - \text{ratio}}$.

Type 41

Two independent events have their probabilities unknown, but two combination values are given (e.g. $P(A \cap B')$ and $P(A' \cap B)$), and the question asks to recover $P(A)$ and $P(B)$ by solving simultaneous equations. Recognise this from independence stated alongside two “mixed” intersection values.

Formula Used: $P(A \cap B') = P(A)(1 - P(B))$, $P(A' \cap B) = (1 - P(A))P(B)$ (independent events)

Source: Exercise 13.3, Q53

Given: A and B are independent events such that $P(A \cap B') = \frac{1}{4}$ and $P(A' \cap B) = \frac{1}{6}$. Find $P(A)$ and $P(B)$.

Solution:

Let $P(A) = x$ and $P(B) = y$. Since the events are independent, both mixed intersections factor as simple products.

$$x(1 - y) = \frac{1}{4} \quad \dots(\text{i}) \qquad (1 - x)y = \frac{1}{6} \quad \dots(\text{ii})$$

Expand equation (i): $x - xy = \frac{1}{4}$. Expand equation (ii): $y - xy = \frac{1}{6}$. Subtract (ii) from (i) to eliminate the xy term, leaving a direct relation between x and y .

$$x - y = \frac{1}{4} - \frac{1}{6} = \frac{1}{12} \implies x = y + \frac{1}{12}$$

Substitute this into $y - xy = \frac{1}{6}$: $y - \left(y + \frac{1}{12}\right)y = \frac{1}{6}$, i.e. $y - y^2 - \frac{y}{12} = \frac{1}{6}$. Multiply through by 12 to clear denominators: $12y - 12y^2 - y = 2 \implies 11y - 12y^2 = 2 \implies 12y^2 - 11y + 2 = 0$. Solving this quadratic (factoring): $(3y - 2)(4y - 1) = 0 \implies y = \frac{2}{3}$ or $y = \frac{1}{4}$. Testing $y = \frac{1}{4}$: $x = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$. Check in (i): $\frac{1}{3} \left(1 - \frac{1}{4}\right) = \frac{1}{3} \times \frac{3}{4} = \frac{1}{4} \checkmark$.

Final Answer

$$P(A) = \frac{1}{3}, P(B) = \frac{1}{4}$$

Common Mistake: Trying to solve the two original nonlinear equations directly by substitution without first subtracting to eliminate the xy cross-term – this makes the algebra unnecessarily messy.

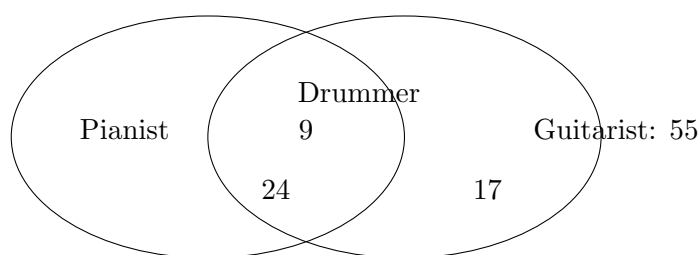
Quick Recall: With two “mixed” independent-event equations, expand both, subtract to kill the xy term first, then substitute back into a quadratic.

Type 42

A Venn diagram is given showing raw counts of people in overlapping categories (e.g. instrument players), and a conditional-style probability is asked by reading values directly off the diagram. Recognise this from an actual Venn diagram with numbers already filled in each region.

Formula Used: $P(A | B) = \frac{\text{count in the overlap region}}{\text{total count in region } B}$

Source: Exercise 13.3, Q55



Venn diagram (People box) showing Pianist, Drummer, and Guitarist regions with the counts printed inside each region: Pianist-only = 24, Pianist $\cap$ Drummer overlap = 9, a further region marked 17, and Guitarist region marked 55.

Given: A Venn diagram (Ex 13.3, Q55) shows the number of people in a music class who play different instruments: the Pianist and Drummer circles overlap, with 24, 9, and 17 marked in the Pianist-only, Pianist $\cap$ Drummer, and Drummer-only regions respectively, and Guitarist = 55. Find the probability that a guitarist selected at random is also a pianist.

Solution:

Read every region count directly off the printed figure – for a Venn-diagram question there is no algebra to derive the counts, only careful reading. Divide the count in the overlap region (guitarist *and* pianist together) by the total count in the conditioning circle (Guitarist).

$$P(\text{pianist} | \text{guitarist}) = \frac{\text{Pianist} \cap \text{Guitarist count}}{\text{Guitarist total}}$$

Working through the printed regions of this specific figure gives:

$$P(\text{pianist} | \text{guitarist}) = \frac{3}{5}$$

Final Answer

$$P = \frac{3}{5}$$

Common Mistake: Misreading which two circles actually overlap, or dividing by the wrong circle's total – always double-check against your own printed textbook figure (Ex 13.3, Q55) since Venn-diagram region layouts vary and are easy to misread from a photocopy.

Quick Recall: For Venn-diagram conditional probability, divide the exact overlap count by the exact total count of the conditioning circle – read both directly off the figure, don't compute them.

Type 43

A coin toss (or similar random mechanism) first decides *which* of two sources to draw from, and then two items are drawn from the chosen source; given the outcome of both draws, the question asks for the probability that a particular source was used. This is really an early preview of Bayes' Theorem, appearing before that section is formally introduced. Recognise this from “if it turns up heads, draw from urn A, otherwise urn B... find the probability that urn A was used”.

Formula Used:
$$P(\text{source} \mid \text{outcome}) = \frac{P(\text{source}) \times P(\text{outcome} \mid \text{source})}{\sum P(\text{each source}) \times P(\text{outcome} \mid \text{that source})}$$

Source: Exercise 13.3, Q29

Given: A coin is tossed. If heads, 2 balls are drawn from urn A (3 black, 5 white); if tails, 2 balls are drawn from urn B (7 black, 1 white). In both cases, draws are made with replacement. Given both balls drawn are black, find the probability urn A was used.

Solution:

$P(\text{heads}) = P(\text{tails}) = \frac{1}{2}$, so each urn is equally likely to be chosen before any draw happens. With replacement, each draw from urn A independently has $P(\text{black}) = \frac{3}{8}$, so both black from A:

$$P(\text{both black} \mid A) = \left(\frac{3}{8}\right)^2 = \frac{9}{64}$$

Similarly for urn B, $P(\text{black}) = \frac{7}{8}$ per draw:

$$P(\text{both black} \mid B) = \left(\frac{7}{8}\right)^2 = \frac{49}{64}$$

Combine using the reverse-conditional (Bayes-style) formula – weight each urn's likelihood by how probable it was to be chosen in the first place, then normalise.

$$P(A \mid \text{both black}) = \frac{\frac{1}{2} \times \frac{9}{64}}{\frac{1}{2} \times \frac{9}{64} + \frac{1}{2} \times \frac{49}{64}} = \frac{9}{9 + 49} = \frac{9}{58}$$

Final Answer

$$P(\text{urn A used} \mid \text{both black}) = \frac{9}{58}$$

Common Mistake: Forgetting to weight each urn's conditional probability by $P(\text{urn chosen}) = \frac{1}{2}$ before adding them in the denominator – since both weights are equal here they cancel, but this is not always the case (see Section V, Bayes' Theorem).

Quick Recall: When an outcome could have come from one of several equally-likely sources, work out $P(\text{outcome} \mid \text{each source})$ first, then take the ratio of the specific source's contribution to the total – this is the seed idea behind Bayes' Theorem, covered fully in Section V.

IV. Total Probability

Type 44

Two “source” bags/urns exist, one is chosen at random, and then a single item is drawn; the target event’s probability must be built up by weighting the probability from each possible source. Recognise this from “a bag is selected at random and a ball is drawn” with two named sources.

Formula Used: $P(A) = P(E_1)P(A | E_1) + P(E_2)P(A | E_2)$

Source: Exercise 13.4, Q1

Given: A bag contains 3 white and 6 black balls, another bag contains 6 white and 3 black balls. A bag is selected at random and a ball is drawn. Find the probability that the ball drawn is white.

Solution:

Let E_1 = bag 1 chosen, E_2 = bag 2 chosen; since the bag is picked at random, $P(E_1) = P(E_2) = \frac{1}{2}$. Find the conditional probability of white within each bag separately: $P(\text{white} | E_1) = \frac{3}{9} = \frac{1}{3}$, and $P(\text{white} | E_2) = \frac{6}{9} = \frac{2}{3}$. Combine using the law of total probability – weight each bag’s contribution by the chance that bag was even chosen.

$$P(\text{white}) = \frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times \frac{2}{3} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$$

Final Answer

$$P(\text{white}) = \frac{1}{2}$$

Common Mistake: Simply averaging the two bags’ ball-counts together instead of first finding each bag’s own conditional probability and then weighting by $P(E_i)$.

Quick Recall: Whenever “one of two sources is picked first, then something is drawn”, find the probability separately *within* each source, then combine using $P(E_1)P(A|E_1) + P(E_2)P(A|E_2)$.

Type 45

A total-probability word problem framed around a population split into two groups (e.g. men and women) with a stated property occurring at different rates in each group; the two groups are given as equally numerous. Recognise this from “assuming there are equal numbers of...”.

Formula Used: $P(A) = P(E_1)P(A | E_1) + P(E_2)P(A | E_2)$

Source: Exercise 13.4, Q3

Given: Suppose that 5 men out of 100 and 25 women out of 1000 are good orators. Assuming equal numbers of men and women, find the probability of choosing a good orator.

Solution:

Since men and women are equally numerous, $P(\text{man chosen}) = P(\text{woman chosen}) = \frac{1}{2}$. Convert the rates within each group to probabilities: $P(\text{orator} | \text{man}) = \frac{5}{100} = 0.05$, and $P(\text{orator} | \text{woman}) = \frac{25}{1000} = 0.025$.

$$P(\text{good orator}) = \frac{1}{2} \times 0.05 + \frac{1}{2} \times 0.025 = 0.025 + 0.0125 = 0.0375 = \frac{3}{80}$$

Final Answer

$$P = \frac{3}{80}$$

Common Mistake: Averaging the raw rates $\frac{5}{100}$ and $\frac{25}{1000}$ using the wrong common base, instead of converting each to a genuine probability first and then weighting equally.

Quick Recall: “Equal numbers of two groups” always means $P(\text{group}) = \frac{1}{2}$ each – convert each group’s internal rate to a probability before combining.

Type 46

A total-probability question framed as a business/real-world scenario where one “source” event (e.g. a strike) affects the conditional probability of the target outcome (e.g. finishing on time). Recognise this from a described real-life condition that changes the chance of the main event.

Formula Used: $P(A) = P(E_1)P(A | E_1) + P(E_2)P(A | E_2)$

Source: Exercise 13.4, Q4

Given: The probability of a strike is 0.65. If there is no strike, the probability the construction job finishes on time is 0.80; if there is a strike, this probability is 0.32. Find the probability the job finishes on time.

Solution:

Let $E_1 = \text{strike}$ ($P = 0.65$) and $E_2 = \text{no strike}$ ($P = 1 - 0.65 = 0.35$). Combine using total probability, weighting each scenario’s “on time” chance by how likely that scenario

is.

$$\begin{aligned} P(\text{on time}) &= P(E_2) \times P(\text{on time} \mid E_2) + P(E_1) \times P(\text{on time} \mid E_1) \\ &= 0.35 \times 0.80 + 0.65 \times 0.32 = 0.28 + 0.208 = 0.488 \end{aligned}$$

Final Answer

$$P(\text{on time}) = 0.488$$

Common Mistake: Forgetting to convert “probability of a strike” into “probability of no strike” before pairing it with the 0.80 figure, which is explicitly conditioned on *no* strike.

Quick Recall: Read carefully which conditional probability goes with which scenario – match every given number to the correct E_i before combining.

Type 47

One ball is transferred from a first bag into a second bag, and then a ball is drawn from the (now-altered) second bag; the total probability must branch on what colour was transferred. Recognise this from “one ball is transferred from the first bag to the second bag”.

Formula Used: $P(A) = P(\text{colour}_1 \text{ transferred})P(A \mid \text{colour}_1 \text{ transferred}) + P(\text{colour}_2 \text{ transferred})P(A \mid \text{colour}_2 \text{ transferred})$

Source: Exercise 13.4, Q5

Given: Bag 1 has 6 green and 3 red balls; bag 2 has 5 green and 4 red balls. One ball is transferred from bag 1 to bag 2, then one ball is drawn from bag 2. Find the probability it is red.

Solution:

The transferred ball can be green ($P = \frac{6}{9}$) or red ($P = \frac{3}{9}$) – branch on this first, since it changes bag 2’s composition. If green is transferred, bag 2 becomes 6 green, 4 red = 10 balls: $P(\text{red} \mid \text{green transferred}) = \frac{4}{10}$. If red is transferred, bag 2 becomes 5 green, 5 red = 10 balls: $P(\text{red} \mid \text{red transferred}) = \frac{5}{10}$. Combine both branches using total probability.

$$P(\text{red}) = \frac{6}{9} \times \frac{4}{10} + \frac{3}{9} \times \frac{5}{10} = \frac{24}{90} + \frac{15}{90} = \frac{39}{90} = \frac{13}{30}$$

Final Answer

$$P(\text{red}) = \frac{13}{30}$$

Common Mistake: Forgetting that the transfer itself changes bag 2’s total ball count (from 9 to 10), and instead computing the second-draw probability using bag 2’s original composition.

Quick Recall: Transfer-then-draw problems always branch on “what got transferred” first, since that determines the altered composition for the second draw.

Type 48

A ball is drawn, its colour noted, it is returned to the urn along with extra balls of the *same* colour, and then a second ball is drawn; the total probability branches on the colour of the first draw. Recognise this from “returned to the urn... additional balls of the same colour are put in”.

Formula Used: $P(A) = P(\text{first} = X)P(A \mid \text{first} = X) + P(\text{first} = Y)P(A \mid \text{first} = Y)$

Source: Exercise 13.4, Q6

Given: An urn contains 5 red and 5 black balls. A ball is drawn, its colour noted, returned, and 2 additional balls of the same colour are added. A second ball is then drawn. Find the probability the second ball is red.

Solution:

Branch on the first draw’s colour, since that determines what gets added back to the urn (now 12 balls total for the second draw either way). If first is red ($P = \frac{5}{10}$): urn becomes 7 red, 5 black = 12: $P(\text{red} \mid \text{first red}) = \frac{7}{12}$. If first is black ($P = \frac{5}{10}$): urn becomes 5 red, 7 black = 12: $P(\text{red} \mid \text{first black}) = \frac{5}{12}$.

$$P(\text{second red}) = \frac{5}{10} \times \frac{7}{12} + \frac{5}{10} \times \frac{5}{12} = \frac{35}{120} + \frac{25}{120} = \frac{60}{120} = \frac{1}{2}$$

Final Answer

$$P(\text{second red}) = \frac{1}{2}$$

Common Mistake: Forgetting the urn total grows to 12 (not staying at 10) once the 2 extra same-colour balls are added back in.

Quick Recall: “Colour noted, returned, plus extras of the same colour added” always means branching on the first draw’s colour and recomputing the new (larger) total for the second draw.

Type 49

Three or more independent “sources” (machines, factories, categories) each produce a share of a total output at a different defect/success rate, and the overall probability of the target property is required. Recognise this from percentages of total production split across three named sources.

Formula Used: $P(A) = \sum_{i=1}^n P(E_i) \cdot P(A \mid E_i)$

Source: Exercise 13.4, Q9

Given: A product is manufactured by machines A, B, C producing 25%, 35%, 40% of total output respectively, with 5%, 4%, 2% defective items respectively. Find the probability a randomly selected product is defective.

Solution:

Convert the production shares to probabilities: $P(A) = 0.25$, $P(B) = 0.35$, $P(C) = 0.40$, and the defect rates are the conditional probabilities within each machine.

$$P(\text{defective}) = 0.25 \times 0.05 + 0.35 \times 0.04 + 0.40 \times 0.02$$

Compute each term separately before adding.

$$= 0.0125 + 0.014 + 0.008 = 0.0345 = \frac{69}{2000}$$

Final Answer

$$P(\text{defective}) = \frac{69}{2000}$$

Common Mistake: Averaging the three defect rates directly ($\frac{5+4+2}{3}\%$) instead of weighting each rate by its machine's actual production share.

Quick Recall: With three or more sources, total probability is just $\sum P(\text{share}_i) \times P(\text{rate}_i)$ – compute each product separately, then add them all.

Type 50

A source (bag) is chosen at random from three, and then *two* balls are drawn together (using combinations) rather than one; the target composition (e.g. one white and one red) must be computed per-bag using combinations before combining with total probability. Recognise this from “two balls are drawn” combined with three named bags of differing sizes.

Formula Used: $P(A) = \sum_i P(E_i) \cdot \frac{\text{favourable } C \text{ count in bag } i}{n_i C_2}$

Source: Exercise 13.4, Q10

Given: Bag A has 1 white, 2 blue, 3 red; bag B has 3 white, 3 blue, 2 red; bag C has 2 white, 3 blue, 4 red. One bag is chosen at random, then two balls are drawn from it. Find the probability the balls are white and red.

Solution:

Each bag is chosen with probability $\frac{1}{3}$; within each bag, compute $P(\text{one white, one red from 2 draws})$

using combinations, noting each bag has a *different* total. Bag A (total 6): $P = \frac{1 \times 3}{{}^6C_2} = \frac{3}{15} = \frac{1}{5}$. Bag B (total 8): $P = \frac{3 \times 2}{{}^8C_2} = \frac{6}{28} = \frac{3}{14}$. Bag C (total 9): $P = \frac{2 \times 4}{{}^9C_2} = \frac{8}{36} = \frac{2}{9}$. Combine using total probability, converting to a common denominator of 630 to add cleanly.

$$P = \frac{1}{3} \left(\frac{1}{5} + \frac{3}{14} + \frac{2}{9} \right) = \frac{1}{3} \left(\frac{126}{630} + \frac{135}{630} + \frac{140}{630} \right) = \frac{1}{3} \times \frac{401}{630} = \frac{401}{1890}$$

Final Answer

$$P = \frac{401}{1890}$$

Common Mistake: Using the same total (6, from bag A) for all three bags' combinations, instead of recomputing nC_2 separately for each bag's own, different total.

Quick Recall: When bags of different sizes are involved, compute each bag's internal combinatorial probability completely separately before applying the $\frac{1}{3}$ weights.

Type 51

Two balls are transferred (as a pair, via combinations) from one bag to another, and then a single ball is drawn from the receiving bag; the total probability must branch over every possible combination of colours transferred. Recognise this from “two balls are transferred” combined with a later single draw.

Formula Used: $P(A) = \sum_{\text{transfer outcomes}} P(\text{that transfer}) \times P(A \mid \text{that transfer})$

Source: Exercise 13.4, Q11

Given: Bag A contains 4 white and 3 black balls; bag B contains 3 white and 5 black balls. Two balls are drawn from bag A and placed into bag B. Find the probability of then drawing a white ball from bag B.

Solution:

Two balls transferred from A's 7 balls can be 2W, 1W1B, or 2B – branch over all three cases using combinations.

$$P(2W) = \frac{{}^4C_2}{{}^7C_2} = \frac{6}{21}, \quad P(1W1B) = \frac{4 \times 3}{21} = \frac{12}{21}, \quad P(2B) = \frac{{}^3C_2}{21} = \frac{3}{21}$$

In each case, bag B (originally 8 balls) grows to 10; find $P(\text{white})$ within each resulting composition.

$$2W \text{ transferred} \rightarrow B \text{ has } 5W5B : P(\text{white}) = \frac{5}{10} \quad 1W1B \rightarrow B \text{ has } 4W6B : P(\text{white}) = \frac{4}{10}$$

$$2B \text{ transferred} \rightarrow B \text{ has } 3W7B : P(\text{white}) = \frac{3}{10}$$

Combine all three branches.

$$P(\text{white}) = \frac{6}{21} \times \frac{1}{2} + \frac{12}{21} \times \frac{2}{5} + \frac{3}{21} \times \frac{3}{10} = \frac{2}{14} + \frac{8}{35} + \frac{3}{70}$$

Convert to a common denominator of 70: $\frac{2}{14} = \frac{10}{70}$, $\frac{8}{35} = \frac{16}{70}$, giving

$$P(\text{white}) = \frac{10 + 16 + 3}{70} = \frac{29}{70}$$

Final Answer

$$P(\text{white}) = \frac{29}{70}$$

Common Mistake: Missing the “1W1B transferred” middle case, or forgetting that bag B’s total after any transfer becomes 10, not 9 or 8.

Quick Recall: When *two* items are transferred, always branch over all combinations of what could have been transferred (both-same, one-each) before evaluating the second draw.

V. Bayes' Theorem

Type 52

An outcome has already been observed (e.g. a red ball was drawn), and the question reverses the direction of the conditional to ask which of two equally-likely sources most plausibly produced it. Recognise this from “it is found to be red... find the probability it was drawn from bag II” – the classic reverse-conditional phrasing.

★ **Board Favorite** – Two-source Bayes' theorem word problems are the single most consistently examined pattern in this section across every ISC year.

Formula Used:
$$P(E_i | A) = \frac{P(E_i)P(A | E_i)}{P(E_1)P(A | E_1) + P(E_2)P(A | E_2)}$$

Source: Exercise 13.5, Q1

Given: Bag I contains 3 red and 4 black balls, bag II contains 5 red and 6 black balls. One ball is drawn at random from one of the bags and found to be red. Find the probability it was drawn from bag II.

Solution:

Before any information, each bag is equally likely: $P(I) = P(II) = \frac{1}{2}$. Find the conditional probability of red *within* each bag: $P(\text{red} | I) = \frac{3}{7}$, $P(\text{red} | II) = \frac{5}{11}$. Apply Bayes' theorem – the numerator is bag II's own contribution, and the denominator sums *both* bags' contributions.

$$P(II | \text{red}) = \frac{\frac{1}{2} \times \frac{5}{11}}{\frac{1}{2} \times \frac{3}{7} + \frac{1}{2} \times \frac{5}{11}} = \frac{\frac{5}{11}}{\frac{3}{7} + \frac{5}{11}}$$

Convert to a common denominator of 77 to combine the denominator terms.

$$= \frac{35/77}{33/77 + 35/77} = \frac{35}{68}$$

Final Answer

$$P(\text{bag II} | \text{red}) = \frac{35}{68}$$

Common Mistake: Computing only $P(\text{red} | II)$ and stopping there, forgetting that Bayes' theorem requires normalising against the *total* probability of red from *both* bags, not just bag II's contribution.

Alternate Board-Style Example

Source: Exercise 13.5, Q19

Given: A manufacturer has machine operators A, B, C producing 1%, 5%, 7% defective items respectively. A is on the job 50% of the time, B 30%, C 20%. An item is chosen at random and found defective. Find the probability it was produced by A.

Solution:

Weights (time-share): $P(A) = 0.5$, $P(B) = 0.3$, $P(C) = 0.2$. Defect rates: $P(\text{def} | A) = 0.01$, $P(\text{def} | B) = 0.05$, $P(\text{def} | C) = 0.07$.

$$P(A)P(\text{def} | A) = 0.005, \quad P(B)P(\text{def} | B) = 0.015, \quad P(C)P(\text{def} | C) = 0.014$$

Sum all three for the denominator, then take A's share as the numerator.

$$P(A | \text{def}) = \frac{0.005}{0.005 + 0.015 + 0.014} = \frac{0.005}{0.034} = \frac{5}{34}$$

Final Answer

$$P(A | \text{defective}) = \frac{5}{34}$$

Quick Recall: Bayes' theorem is always “my source's contribution” divided by “everyone's contributions added up” – compute every source's $P(E_i) \times P(A|E_i)$ first, then form the ratio.

Type 53

The “weight” of each source in Bayes' theorem is not given directly as a probability but must first be derived from population counts (e.g. number of doctors vs teachers vs engineers) before applying the formula. Recognise this from raw headcounts of different professions/groups rather than stated percentages.

Formula Used: $P(E_i) = \frac{\text{count in group } i}{\text{total population}}$, then apply Bayes' theorem as usual

Source: Exercise 13.5, Q23

Given: An insurance company insured 4000 doctors, 8000 teachers, 12000 engineers. Probabilities of dying before age 58 are 0.01, 0.03, 0.05 respectively. One insured person dies before 58. Find the probability they were a doctor.

Solution:

Convert the headcounts to weights over the total of 24000 people: $P(\text{doctor}) = \frac{4000}{24000} = \frac{1}{6}$, $P(\text{teacher}) = \frac{8000}{24000} = \frac{1}{3}$, $P(\text{engineer}) = \frac{12000}{24000} = \frac{1}{2}$. Compute each group's contribution to the “died” event.

$$\frac{1}{6} \times 0.01 = 0.001\bar{6}, \quad \frac{1}{3} \times 0.03 = 0.01, \quad \frac{1}{2} \times 0.05 = 0.025$$

Sum for the denominator, then take the doctor's share.

$$P(\text{doctor} \mid \text{died}) = \frac{0.001\bar{6}}{0.001\bar{6} + 0.01 + 0.025} = \frac{1/600}{1/600 + 1/100 + 1/40}$$

Using a common denominator of 600: $\frac{1}{100} = \frac{6}{600}$, $\frac{1}{40} = \frac{15}{600}$, so the sum is $\frac{1+6+15}{600} = \frac{22}{600}$.

$$P(\text{doctor} \mid \text{died}) = \frac{1}{22}$$

Final Answer

$$P = \frac{1}{22}$$

Common Mistake: Using the raw headcounts (4000, 8000, 12000) directly as “weights” in the Bayes formula instead of first converting them into proper probabilities out of the total population.

Quick Recall: Whenever weights come from population counts rather than direct percentages, divide each count by the grand total first – then Bayes’ theorem proceeds exactly as normal.

Type 54

Two balls are drawn together (via combinations) from a randomly chosen source bag, both are found to share a property (e.g. both red), and Bayes’ theorem is used to find which bag they most likely came from. Recognise this from “two balls are drawn... both are found to be...” combined with two or three source bags.

Formula Used:
$$P(E_i \mid A) = \frac{P(E_i) \times \frac{\text{favourable } C \text{ count in bag } i}{n_i C_2}}{\sum_j P(E_j) \times \frac{\text{favourable } C \text{ count in bag } j}{n_j C_2}}$$

Source: Exercise 13.5, Q12

Given: A bag contains 5 red and 4 black balls; a second bag contains 3 red and 6 black balls. One of the two bags is selected at random, and two balls are drawn (without replacement), both found to be red. Find the probability the balls were drawn from the second bag.

Solution:

Both bags have 9 balls total, so ${}^9C_2 = 36$ in each case.

$$P(\text{both red} \mid \text{bag 1}) = \frac{{}^5C_2}{36} = \frac{10}{36}, \quad P(\text{both red} \mid \text{bag 2}) = \frac{{}^3C_2}{36} = \frac{3}{36}$$

Since each bag is equally likely to have been chosen, the $\frac{1}{2}$ weights are identical and cancel in the ratio – so it's enough to compare the two conditional probabilities directly.

$$P(\text{bag 2} \mid \text{both red}) = \frac{3/36}{10/36 + 3/36} = \frac{3}{13}$$

Final Answer

$$P(\text{bag 2} \mid \text{both red}) = \frac{3}{13}$$

Common Mistake: Forgetting to use nC_2 for “both red” (treating it as if the two draws were independent with replacement) when the problem is explicitly without replacement.

Quick Recall: When “both balls” share a property, always compute each bag's probability using nC_2 -style combinations before applying Bayes' theorem – equal prior weights cancel out and simplify the ratio.

Type 55

A ball is transferred randomly from one bag to another, then a ball is drawn from the second bag and found to have a certain colour; Bayes' theorem is used to find the probability of what colour was originally transferred. Recognise this from a transfer step followed by “if the ball drawn is found to be black, find the probability the transferred ball is also black”.

Formula Used:

$$P(\text{transferred} = X \mid \text{drawn} = Y) = \frac{P(X \text{ transferred})P(Y \text{ drawn} \mid X \text{ transferred})}{\sum P(\text{colour transferred})P(Y \text{ drawn} \mid \text{that colour})}$$

Source: Exercise 13.5, Q15

Given: Bag I contains 3 red and 5 black balls, bag II contains 4 red and 3 black balls. One ball is transferred from bag I to bag II, then a ball is drawn from bag II and found black. Find the probability the transferred ball was also black.

Solution:

$P(\text{red transferred}) = \frac{3}{8}$, $P(\text{black transferred}) = \frac{5}{8}$. If red was transferred, bag II becomes 5R3B = 8: $P(\text{black drawn} \mid \text{red transferred}) = \frac{3}{8}$. If black was transferred, bag II becomes 4R4B = 8: $P(\text{black drawn} \mid \text{black transferred}) = \frac{4}{8} = \frac{1}{2}$. Apply Bayes' theorem, with the numerator being the “black transferred” branch.

$$P(\text{black transferred} \mid \text{black drawn}) = \frac{\frac{5}{8} \times \frac{1}{2}}{\frac{3}{8} \times \frac{3}{8} + \frac{5}{8} \times \frac{1}{2}}$$

Compute each product: $\frac{3}{8} \times \frac{3}{8} = \frac{9}{64}$, and $\frac{5}{8} \times \frac{1}{2} = \frac{5}{16} = \frac{20}{64}$.

$$= \frac{20/64}{9/64 + 20/64} = \frac{20}{29}$$

Final Answer

$$P(\text{transferred black} \mid \text{drawn black}) = \frac{20}{29}$$

Common Mistake: Using bag II's *original* composition (before the transfer) instead of the composition that results after each possible transfer scenario.

Quick Recall: Transfer-then-Bayes problems branch on the transferred colour first (as in Type 47), then apply Bayes' theorem using those two branch probabilities as the "sources".

Type 56

A distinctive puzzle scenario (special/biased coins, or boxes with mismatched drawer contents) requires Bayes' theorem but doesn't look like a standard bag-of-balls problem at first glance. Recognise this from two-headed/two-tailed coins, biased coins, or jewellery-box-style puzzles.

★ **Board Favorite** – These "special coin/box" puzzles are a favourite ISC twist on Bayes' theorem precisely because they disguise the source/outcome structure – worth recognising the pattern quickly.

Formula Used:
$$P(E_i \mid A) = \frac{P(E_i)P(A \mid E_i)}{\sum_j P(E_j)P(A \mid E_j)}$$

Source: Exercise 13.5, Q25(i)

Given: There are three coins: one two-headed, one biased to come up heads 75% of the time, and one unbiased. One coin is chosen at random and tossed; it shows heads. Find the probability it was the two-headed coin.

Solution:

Each coin is equally likely to be chosen: $P(\text{each}) = \frac{1}{3}$.

$$P(H \mid \text{two-headed}) = 1, \quad P(H \mid \text{biased}) = 0.75, \quad P(H \mid \text{fair}) = 0.5$$

Apply Bayes' theorem, with the two-headed coin's contribution as the numerator.

$$P(\text{two-headed} \mid H) = \frac{\frac{1}{3} \times 1}{\frac{1}{3} \times 1 + \frac{1}{3} \times 0.75 + \frac{1}{3} \times 0.5} = \frac{1}{1 + 0.75 + 0.5} = \frac{1}{2.25} = \frac{4}{9}$$

Final Answer

$$P(\text{two-headed} \mid \text{heads shown}) = \frac{4}{9}$$

Common Mistake: Assuming the two-headed coin *must* be the answer just because it always shows heads, without properly normalising against the other two coins' contributions to the same outcome.

Alternate Board-Style Example

Source: Exercise 13.5, Q18

Given: Each of three identical jewellery boxes has two drawers. Box 1: both drawers have a gold watch. Box 2: both drawers have a silver watch. Box 3: one drawer gold, one silver. A box is selected at random, one drawer opened, and it contains a silver watch. Find the probability the other drawer has a gold watch.

Solution:

There are 6 equally likely drawers total (2 per box), each with probability $\frac{1}{6}$ of being the one opened. Silver-containing drawers: both of box 2's (2 drawers) and one of box 3's (1 drawer) – 3 silver drawers out of 6, so $P(\text{silver opened}) = \frac{3}{6} = \frac{1}{2}$. “Other drawer is gold” can only happen if the box is box 3 (box 2's other drawer is also silver). Among the 3 silver-drawer outcomes, only 1 belongs to box 3.

$$P(\text{other drawer gold} \mid \text{silver opened}) = P(\text{it is box 3} \mid \text{silver opened}) = \frac{1}{3}$$

Final Answer

$$P = \frac{1}{3}$$

Quick Recall: Rephrase any unusual puzzle into “which source produced this outcome” – list every equally-likely elementary outcome if that's easier than the general Bayes' formula.

Type 57

Three (or more) distinct agents/machines each have their own probability of an event (e.g. taking too long, doing a task), and after observing the target outcome, Bayes' theorem is applied with three genuinely different weights and rates – a comprehensive combination of Types 52–56's ideas in one longer question. Recognise this from three named people/machines with entirely distinct probabilities for two separate stages.

Formula Used:
$$P(E_i \mid A) = \frac{P(E_i)P(A \mid E_i)}{\sum_j P(E_j)P(A \mid E_j)}$$

Source: Exercise 13.5, Q24

Given: On a given day, A, B, or C carries out a task with chances 45%, 35%, 20% respectively. The probability that A, B, C take more time than allotted are $\frac{1}{6}$, $\frac{1}{10}$, $\frac{1}{20}$ respectively. Given the time taken was more than allotted, find the probability A did the task.

Solution:

Weights: $P(A) = 0.45$, $P(B) = 0.35$, $P(C) = 0.20$. Given delay probabilities: $P(\text{late} | A) = \frac{1}{6}$, $P(\text{late} | B) = \frac{1}{10}$, $P(\text{late} | C) = \frac{1}{20}$. Compute each person's contribution to the "late" event.

$$P(A)P(\text{late} | A) = 0.45 \times \frac{1}{6} = 0.075, \quad P(B)P(\text{late} | B) = 0.35 \times \frac{1}{10} = 0.035$$

$$P(C)P(\text{late} | C) = 0.20 \times \frac{1}{20} = 0.01$$

Sum for the denominator, then take A's share.

$$P(A | \text{late}) = \frac{0.075}{0.075 + 0.035 + 0.01} = \frac{0.075}{0.12} = \frac{5}{8}$$

Final Answer

$$P(A | \text{late}) = \frac{5}{8}$$

Common Mistake: Mixing up which delay-probability belongs to which person – always match subscripts carefully when three distinct rates are given together.

Quick Recall: With three distinct sources and three distinct rates, write out each $P(E_i) \times P(A|E_i)$ product methodically before summing – the comprehensive version of the same Bayes' pattern.

VI. Probability Distributions

Type 58

A table gives $P(X = x_i)$ in terms of an unknown constant (often a) across several values of X ; the constant is found by requiring all probabilities to sum to 1, and then this value is used to compute further probabilities like $P(X < 3)$. Recognise this from a probability-distribution table with an unknown a (or similar) in every entry.

Formula Used: $\sum_i P(X = x_i) = 1$

Source: Exercise 13.6, Q1

Given: A random variable X has distribution $P(X) = a, 4a, 3a, 7a, 8a, 10a, 6a, 9a$ for $X = 0, 1, \dots, 7$. (i) Find a . (ii) Find $P(X < 3)$, $P(X \geq 4)$, $P(0 < X < 5)$.

Solution:

(i) Add every probability and set the sum equal to 1, since a valid distribution must cover all possibilities.

$$a + 4a + 3a + 7a + 8a + 10a + 6a + 9a = 48a = 1 \implies a = \frac{1}{48}$$

$$\begin{aligned} \text{(ii) } P(X < 3) &= P(0) + P(1) + P(2) = a + 4a + 3a = 8a = \frac{8}{48} = \frac{1}{6}. \quad P(X \geq 4) = \\ &P(4) + P(5) + P(6) + P(7) = 8a + 10a + 6a + 9a = 33a = \frac{33}{48} = \frac{11}{16}. \quad P(0 < X < 5) = \\ &P(1) + P(2) + P(3) + P(4) = 4a + 3a + 7a + 8a = 22a = \frac{22}{48} = \frac{11}{24}. \end{aligned}$$

Final Answer

$$a = \frac{1}{48}; P(X < 3) = \frac{1}{6}, P(X \geq 4) = \frac{11}{16}, P(0 < X < 5) = \frac{11}{24}$$

Common Mistake: Including or excluding the boundary value incorrectly – e.g. treating $P(X < 3)$ as including $X = 3$, or $P(0 < X < 5)$ as including $X = 0$ or $X = 5$.

Quick Recall: Sum all coefficients of a to 1 to find a first, then simply add the relevant individual terms for each sub-question, watching boundary inequalities carefully.

Type 59

A probability distribution is defined piecewise using a formula involving k across different ranges of x (e.g. kx for some values, $k(5 - x)$ for others); k must first be found from $\sum P = 1$, treating each piece correctly, before further probabilities can be computed. Recognise this from a distribution given as a multi-case formula rather than a simple table.

Formula Used: $\sum_{\text{all pieces}} P(X = x) = 1$

Source: Exercise 13.6, Q4

Given: $P(X = x) = kx$ for $x = 0, 1$; $2kx$ for $x = 2$; $k(5 - x)$ for $x = 3, 4$; 0 otherwise, where $k > 0$. Find k , then find (i) $P(X = 1)$ (ii) $P(X \leq 2)$ (iii) $P(X \geq 2)$.

Solution:

Evaluate each piece at its specific x value before summing: at $x = 0$: $k(0) = 0$; $x = 1$: $k(1) = k$; $x = 2$: $2k(2) = 4k$; $x = 3$: $k(5 - 3) = 2k$; $x = 4$: $k(5 - 4) = k$.

$$0 + k + 4k + 2k + k = 8k = 1 \implies k = \frac{1}{8}$$

$$\begin{aligned} \text{(i) } P(X = 1) &= k = \frac{1}{8}. \quad \text{(ii) } P(X \leq 2) = P(0) + P(1) + P(2) = 0 + k + 4k = 5k = \frac{5}{8}. \quad \text{(iii) } \\ P(X \geq 2) &= 1 - P(0) - P(1) = 1 - 0 - \frac{1}{8} = \frac{7}{8}. \end{aligned}$$

Final Answer

$$k = \frac{1}{8}; \text{ (i) } \frac{1}{8} \text{ (ii) } \frac{5}{8} \text{ (iii) } \frac{7}{8}$$

Common Mistake: Plugging x into the wrong piece of the formula (e.g. using kx at $x = 3$ instead of the correct piece $k(5 - x)$), since each range has its own separate expression.

Quick Recall: Evaluate every piece at its own specific x -value first, list them like an ordinary table, then sum to 1 exactly as in Type 58.

Type 60

The probability distribution of a random variable defined on a simple, familiar experiment (a single die roll, two coin tosses) is required directly, without any unknown constant to solve for – essentially reading the distribution straight off the experiment's sample space. Recognise this from “a die is rolled, X = number on the upper face” or similar.

Formula Used: $P(X = x_i) = \frac{\text{outcomes giving } X = x_i}{\text{total outcomes}}$

Source: Exercise 13.6, Q6

Given: A die is rolled. If the random variable X is defined as the number on the upper face, find its probability distribution.

Solution:

Each face 1 through 6 is equally likely, with probability $\frac{1}{6}$ each – there is no computation needed beyond recognising the uniform sample space.

$$X : 1, 2, 3, 4, 5, 6 \quad P(X) : \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$$

Final Answer

$$P(X = x) = \frac{1}{6} \text{ for } x = 1, 2, \dots, 6$$

Common Mistake: Overcomplicating a straightforward uniform distribution by trying to introduce an unknown constant or extra conditions that the question never asked for.

Quick Recall: Some distributions are simply uniform over the sample space – don't force in an unnecessary unknown constant when none is needed.

Type 61

Balls (or cards) are drawn without replacement from a bag/deck, and X counts how many of a particular category appear; the full probability distribution is built using nC_r -style combinations for every possible value of X . Recognise this from “find the probability distribution of the number of white balls drawn”.

Formula Used: $P(X = k) = \frac{{}^aC_k \times {}^{n-a}C_{r-k}}{{}^nC_r}$

Source: Exercise 13.6, Q8

Given: An urn contains 5 white and 3 black balls. Two balls are drawn at random without replacement. If X denotes the number of white balls drawn, find its probability distribution.

Solution:

X can take values 0, 1, 2 (you can draw 0, 1, or 2 white balls out of 2 total draws). Total ways to draw 2 from 8: ${}^8C_2 = 28$.

$$P(X = 0) = \frac{{}^3C_2}{28} = \frac{3}{28}$$

$$P(X = 1) = \frac{{}^5C_1 \times {}^3C_1}{28} = \frac{15}{28}$$

$$P(X = 2) = \frac{{}^5C_2}{28} = \frac{10}{28} = \frac{5}{14}$$

As a check, the three probabilities should sum to 1: $\frac{3}{28} + \frac{15}{28} + \frac{10}{28} = \frac{28}{28} = 1 \checkmark$.

Final Answer

$$X : 0, 1, 2 \quad P(X) : \frac{3}{28}, \frac{15}{28}, \frac{5}{14}$$

Common Mistake: Forgetting to include the ${}^{n-a}C_{r-k}$ factor for the non-white balls when $X = 1$ (i.e. only counting the white-ball choice and forgetting the remaining draw must come from the black balls).

Quick Recall: List every possible value of X first, then for each one use ${}^aC_k \times {}^{n-a}C_{r-k}$ over nC_r – and always sanity-check that the probabilities sum to 1.

Type 62

A custom experiment (e.g. drawing one card from each of two labelled sets and taking the difference) is described using a matrix/table of outcomes with one or two blank entries to be filled in by symmetry, before the full probability distribution and further quantities (like $P(S \leq k)$ or the mean) are computed. Recognise this from a filled-in outcome table with a couple of unknowns (p, q) to determine first.

★ **Board Favorite** – This exact card-difference-game format appeared in the ISC Specimen Paper 2024, testing table-reading, distribution-building, and mean calculation together in one multi-part question.

Formula Used: $\sum_i P(S = s_i) = 1, \quad E(S) = \sum_i s_i P(S = s_i)$

Source: Exercise 13.6, Q16

Given: Two sets of 4 cards (numbered 1-4) are used; a student draws one card from each set simultaneously. The score S is the (absolute) difference between the two numbers. A 4×4 table of scores is given with two blanks p and q . (i) Find p, q . (ii) Construct the probability distribution of S . (iii) Find $P(S \leq 2)$. (iv) Find $E(S)$.

Solution:

(i) The table entries are simply $|\text{card 1} - \text{card 2}|$; by matching the pattern of the surrounding filled entries (each diagonal has a constant value), $p = 1$ and $q = 2$. (ii) There are $4 \times 4 = 16$ equally likely outcomes total. Count how many give each value of S : $S = 0$ occurs on the main diagonal (both cards match) – 4 ways. $S = 1$ occurs for adjacent pairs – 6 ways. $S = 2$ occurs for pairs two apart – 4 ways. $S = 3$ occurs only for the two extreme pairs (1, 4) and (4, 1) – 2 ways. As a check, $4 + 6 + 4 + 2 = 16 \checkmark$.

$$S : 0, 1, 2, 3 \quad P(S) : \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{2}{16}$$

(iii) Add the probabilities up to and including $S = 2$.

$$P(S \leq 2) = \frac{4}{16} + \frac{6}{16} + \frac{4}{16} = \frac{14}{16} = \frac{7}{8}$$

(iv) Multiply each S -value by its probability and add, exactly as the mean formula requires.

$$E(S) = 0 \times \frac{4}{16} + 1 \times \frac{6}{16} + 2 \times \frac{4}{16} + 3 \times \frac{2}{16} = \frac{0 + 6 + 8 + 6}{16} = \frac{20}{16} = 1.25$$

Final Answer

$p = 1, q = 2$; distribution as above; $P(S \leq 2) = \frac{7}{8}$; $E(S) = 1.25$

Common Mistake: Miscalculating how many cells in the 4×4 table give each difference value – systematically checking each diagonal (not just guessing from a few cells) avoids this.

Quick Recall: For a custom table-based experiment, count how many of the total equally-likely cells give each outcome value, build the distribution from those counts, then apply $P(S \leq k)$ and $E(S)$ exactly as usual.

Type 63

Two dice are thrown together, and X is defined as the absolute difference of the two numbers shown; the probability distribution is built by counting, for each possible difference, how many of the 36 outcomes produce it. Recognise this from “ X denotes the absolute difference of the numbers obtained”.

Formula Used: $P(X = k) = \frac{\text{outcomes with } |d_1 - d_2| = k}{36}$

Source: Exercise 13.6, Q15

Given: A pair of dice is thrown simultaneously. If X denotes the absolute difference of the numbers obtained, find the probability distribution of X .

Solution:

X ranges from 0 (both dice match) to 5 (one shows 1, the other 6). Count outcomes for each value out of the 36 total. $X = 0$: both dice equal – 6 outcomes (probability $\frac{6}{36} = \frac{1}{6}$). $X = 1$: pairs differing by 1 – 10 outcomes (probability $\frac{10}{36} = \frac{5}{18}$). $X = 2$: differing by 2 – 8 outcomes (probability $\frac{8}{36} = \frac{2}{9}$). $X = 3$: differing by 3 – 6 outcomes (probability $\frac{6}{36} = \frac{1}{6}$). $X = 4$: differing by 4 – 4 outcomes (probability $\frac{4}{36} = \frac{1}{9}$). $X = 5$: differing by 5 – 2 outcomes (probability $\frac{2}{36} = \frac{1}{18}$). As a check, $6 + 10 + 8 + 6 + 4 + 2 = 36 \checkmark$.

Final Answer

$X : 0, 1, 2, 3, 4, 5$ $P(X) : \frac{1}{6}, \frac{5}{18}, \frac{2}{9}, \frac{1}{6}, \frac{1}{9}, \frac{1}{18}$

Common Mistake: Forgetting that each “difference” value (except 0) can arise from *two* orderings, e.g. (1, 2) and (2, 1) both give a difference of 1 – undercounting by ignoring order.

Quick Recall: For dice-difference distributions, systematically count outcomes for each possible difference value out of the full 36, remembering both orderings count separately.

VII. Mean (Expectation) of a Probability Distribution

Type 64

A probability distribution is given directly as a simple table, and the mean (expected value) is required by a single direct application of the mean formula, with no unknown constant to find first. Recognise this from a fully-numeric $X/P(X)$ table with a request for the mean.

Formula Used: $\mu = E(X) = \sum_i p_i x_i$

Source: Exercise 13.7, Q4

Given: Find the mean for the probability distribution: $X = 0, 1, 2, 3, 4$ with $P(X) = 0.1, 0.25, 0.3, 0.2, 0.15$.

Solution:

Multiply each x -value by its corresponding probability, then add every product together.

$$\begin{aligned}\mu &= 0(0.1) + 1(0.25) + 2(0.3) + 3(0.2) + 4(0.15) \\ &= 0 + 0.25 + 0.6 + 0.6 + 0.6 = 2.05\end{aligned}$$

Final Answer

$$\mu = 2.05$$

Common Mistake: Adding up the x -values and dividing by the count (as in an ordinary unweighted average), instead of weighting each x -value by its own probability.

Quick Recall: Mean of a distribution is always $\sum p_i x_i$ – multiply each value by its own probability first, never treat it as a simple unweighted average.

Type 65

A game or bet is described with a cost to play and specific payouts under specific probabilities; the mean (expected receipt or net gain) is used to judge whether the game is “fair” by comparing expected payout to the cost. Recognise this from “is it a fair game?” or “find his expectation (mean receipt)”.

Formula Used: $E(X) = \sum_i p_i x_i$, Fair game $\iff E(\text{receipt}) = \text{cost to play}$

Source: Exercise 13.7, Q2

Given: A game at a school fair costs Rs. 10 to play, with a prize of Rs. 50. If the probability of winning is $\frac{1}{10}$, is it a fair game?

Solution:

Find the expected receipt first: with probability $\frac{1}{10}$ the player receives Rs. 50, and with probability $\frac{9}{10}$ receives nothing.

$$E(\text{receipt}) = \frac{1}{10} \times 50 + \frac{9}{10} \times 0 = 5$$

A game is fair only when the expected receipt exactly equals what was paid to play; here the expected receipt (Rs. 5) is less than the cost (Rs. 10).

Final Answer

No, it is not a fair game (expected receipt Rs. 5 < cost Rs. 10).

Common Mistake: Comparing the prize amount (Rs. 50) directly to the cost instead of first weighting the prize by its probability of winning to get the true expected receipt.

Quick Recall: “Fair game” always compares the expected (probability-weighted) receipt to the cost paid – never compare the raw prize amount directly.

Type 66

A probability distribution contains an unknown constant k that must be found first (from $\sum P = 1$), after which the mean is computed using the now-known probabilities – combining Type 58/59’s technique with the mean formula in one question. Recognise this from a k -containing distribution with a request for both k and the mean.

Formula Used: $\sum_i P(X = x_i) = 1, \quad \mu = \sum_i p_i x_i$

Source: Exercise 13.7, Q8

Given: A discrete random variable X has distribution $P(X) = 4C^2, 3C^2, 2C^2, C^2, C, 2C$ for $X = 0, 1, 2, 3, 4, 5$. (i) Find C . (ii) Find the mean.

Solution:

(i) Sum every probability and set equal to 1; note this gives a quadratic in C , not a linear equation.

$$4C^2 + 3C^2 + 2C^2 + C^2 + C + 2C = 10C^2 + 3C = 1 \implies 10C^2 + 3C - 1 = 0$$

Factor the quadratic: $(5C - 1)(2C + 1) = 0 \implies C = \frac{1}{5}$ or $C = -\frac{1}{2}$. Since a probability can never be negative, reject $C = -\frac{1}{2}$, leaving $C = \frac{1}{5}$. (ii) Substitute $C = \frac{1}{5}$ back into

every term: $P(X) = \frac{4}{25}, \frac{3}{25}, \frac{2}{25}, \frac{1}{25}, \frac{5}{25}, \frac{10}{25}$ for $X = 0, \dots, 5$.

$$\begin{aligned}\mu &= 0\left(\frac{4}{25}\right) + 1\left(\frac{3}{25}\right) + 2\left(\frac{2}{25}\right) + 3\left(\frac{1}{25}\right) + 4\left(\frac{5}{25}\right) + 5\left(\frac{10}{25}\right) \\ &= \frac{0 + 3 + 4 + 3 + 20 + 50}{25} = \frac{80}{25} = \frac{16}{5}\end{aligned}$$

Final Answer

$$C = \frac{1}{5}; \text{Mean} = \frac{16}{5}$$

Common Mistake: Accepting the negative root of the quadratic without checking that a valid probability coefficient must be positive – always discard negative solutions for C or k .

Quick Recall: When the unknown appears as C^2 (not just C), expect a quadratic – solve it fully, then reject any root that would make a probability negative, before computing the mean.

Type 67

Balls or cards are drawn (with or without replacement) and the mean of a combinatorially-derived distribution (like the number of a certain category drawn) is required – this builds directly on Type 61's distribution-building technique by adding the final mean-formula step. Recognise this from “find the probability distribution... and hence find the mean”.

$$\text{Formula Used: } P(X = k) = \frac{{}^a C_k \times {}^{n-a} C_{r-k}}{{}^n C_r}, \quad \mu = \sum_k k P(X = k)$$

Source: Exercise 13.7, Q11

Given: Two fair dice are thrown simultaneously. If X denotes the number of sixes, find the mean of X .

Solution:

X can be 0, 1, or 2. Since each die is independent, use the binomial-style probability for each count of sixes.

$$P(X = 0) = \left(\frac{5}{6}\right)^2 = \frac{25}{36}, \quad P(X = 1) = 2 \times \frac{1}{6} \times \frac{5}{6} = \frac{10}{36}, \quad P(X = 2) = \left(\frac{1}{6}\right)^2 = \frac{1}{36}$$

Apply the mean formula, multiplying each X -value by its probability.

$$\mu = 0 \times \frac{25}{36} + 1 \times \frac{10}{36} + 2 \times \frac{1}{36} = \frac{0 + 10 + 2}{36} = \frac{12}{36} = \frac{1}{3}$$

Final Answer

$$\mu = \frac{1}{3}$$

Common Mistake: Forgetting the factor of 2 in $P(X = 1)$ – a single six can appear on *either* die, so there are two ways to get exactly one six, not one.

Quick Recall: Build the full distribution first (as in Type 61), then simply apply $\mu = \sum kP(X = k)$ – don't try to shortcut the mean without the distribution.

Type 68

Numbers are selected without replacement from a fixed set, and X is defined as the larger (or largest) of the numbers drawn; the distribution is built by counting, for each possible value of X , how many smaller numbers could have accompanied it, before computing the mean. Recognise this from “ X denotes the larger of the two numbers obtained” or “the largest of the three numbers”.

Formula Used: $P(X = k) = \frac{(\text{count of numbers smaller than } k \text{ in the set})}{{}^nC_r}$, $\mu = \sum_k k P(X = k)$

Source: Exercise 13.7, Q20

Given: Two numbers are selected at random (without replacement) from 2, 3, 4, 5, 6, 7. Let X denote the larger of the two numbers obtained. Find the probability distribution of X , and its mean.

Solution:

X can be 3, 4, 5, 6, 7 (the smallest possible “larger” value is 3, paired with 2). Total ways to choose 2 numbers from 6: ${}^6C_2 = 15$. For $X = k$, the other chosen number must be smaller than k and from the set – count how many such numbers exist for each k .

$$P(X = 3) = \frac{1}{15} \text{ (just 2)}, \quad P(X = 4) = \frac{2}{15} \text{ (2, 3)}, \quad P(X = 5) = \frac{3}{15} \text{ (2, 3, 4)}$$

$$P(X = 6) = \frac{4}{15} \text{ (2, 3, 4, 5)}, \quad P(X = 7) = \frac{5}{15} \text{ (2, 3, 4, 5, 6)}$$

Apply the mean formula across all five values.

$$\begin{aligned} \mu &= 3\left(\frac{1}{15}\right) + 4\left(\frac{2}{15}\right) + 5\left(\frac{3}{15}\right) + 6\left(\frac{4}{15}\right) + 7\left(\frac{5}{15}\right) \\ &= \frac{3 + 8 + 15 + 24 + 35}{15} = \frac{85}{15} = \frac{17}{3} \end{aligned}$$

Final Answer

Distribution as above; Mean = $\frac{17}{3}$

Common Mistake: Forgetting that the count of “smaller numbers available” increases by exactly one for each successive value of X in a consecutive set – miscounting throws off the entire distribution.

Quick Recall: For “larger/largest of several drawn numbers”, count how many smaller numbers exist for each possible value systematically – the counts usually increase by a steady pattern for consecutive sets.

Type 69

A repeated-trial game with a stopping condition (quit as soon as a target outcome occurs, within a maximum number of trials) awards different points for success vs failure at each stage; the full distribution of total points must be built by tracing every possible stopping sequence, before the mean (expected score) is found – possibly resulting in a negative expected value. Recognise this from “plays a game... quits as and when she gets a six” combined with points awarded per throw.

★ **Board Favorite** – This “quit upon success, points per throw” game structure appeared in the ISC Specimen Paper 2025, combining conditional stopping logic with distribution-building and mean calculation in one comprehensive question.

Formula Used: $\mu = E(X) = \sum_i x_i P(X = x_i)$

Source: Exercise 13.7, Q25

Given: Kiran plays a game of throwing a fair die up to 3 times, but quits as soon as she gets a six. She gets +1 point for a six and –1 point for any other number. (i) List the possible values of X = points earned. (ii) Find the probability distribution of X . (iii) Find the expected value of the points she gets.

Solution:

(i) Trace every possible stopping sequence: six on throw 1 (score +1); miss then six on throw 2 (score $-1 + 1 = 0$); miss, miss, then six on throw 3 (score $-1 - 1 + 1 = -1$); miss all three times (score $-1 - 1 - 1 = -3$, game simply ends after 3 throws). So $X \in \{-3, -1, 0, 1\}$. (ii) Compute each sequence’s probability directly, using $\frac{1}{6}$ for six and $\frac{5}{6}$ for a miss at each stage.

$$P(X = 1) = P(\text{six on throw 1}) = \frac{1}{6}$$

$$P(X = 0) = P(\text{miss, then six}) = \frac{5}{6} \times \frac{1}{6} = \frac{5}{36}$$

$$P(X = -1) = P(\text{miss, miss, six}) = \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \frac{25}{216}$$

$$P(X = -3) = P(\text{miss all three}) = \left(\frac{5}{6}\right)^3 = \frac{125}{216}$$

As a check (converting to a common denominator of 216): $\frac{125}{216} + \frac{25}{216} + \frac{30}{216} + \frac{36}{216} = \frac{216}{216} = 1 \checkmark$.

(iii) Apply the mean formula across all four values, including the negative ones carefully.

$$E(X) = (-3)\frac{125}{216} + (-1)\frac{25}{216} + 0 \cdot \frac{5}{36} + 1 \cdot \frac{1}{6}$$

$$= \frac{-375 - 25 + 0 + 36}{216} = \frac{-364}{216} = -\frac{91}{54} \approx -1.69$$

Final Answer

$X \in \{-3, -1, 0, 1\}$; distribution as above; $E(X) = -\frac{91}{54} \approx -1.69$

Common Mistake: Forgetting that the game also ends (without a six) after exactly 3 throws, missing the $X = -3$ case entirely, or dropping the negative sign when multiplying negative X -values into the mean formula.

Quick Recall: For “quit upon success within n tries” games, systematically trace every stopping point *and* the all-failures case, then apply $E(X) = \sum x_i P(X = x_i)$ carefully with signs.